

Business Process Automation with VBA and Python

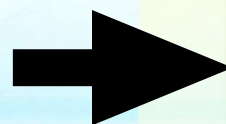
Mr. Eddie Chow / 16 May 2026



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Final Group Presentation



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9. Scraping Tools and Technologies
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11. Introduction to HTML and CSS
12. BeautifulSoup for Web Scraping

Business Process Automation with Python

Python – Origin

- Released in 1991 (same as Visual Basic)
- Free and open-source



Business Process Automation with Python

Motivation – Soft Limits

- Protective measures not matching modern days

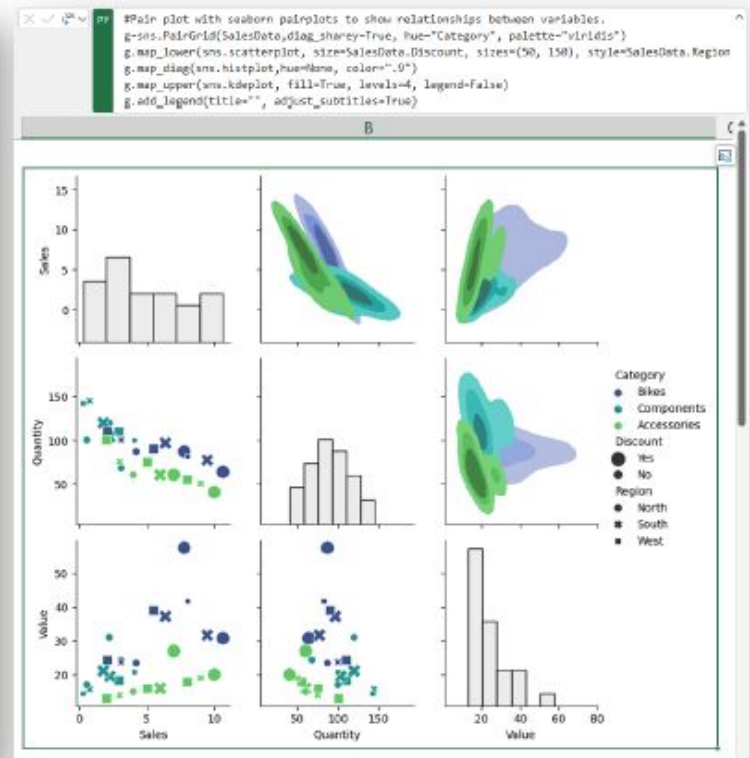
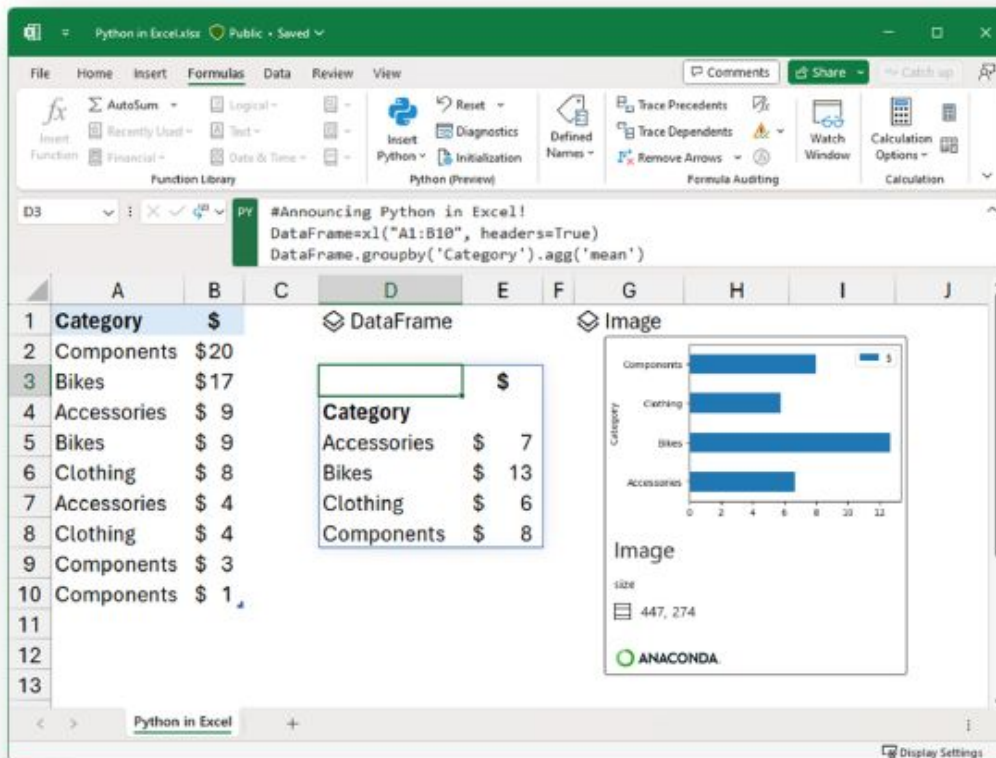
Feature	Maximum limit
Total number of rows and columns on a worksheet	1,048,576 rows by 16,384 columns
Column width	255 characters
Row height	409 points
Page breaks	1,026 horizontal and vertical

1. Microsoft. Excel Specifications and limits. <https://support.microsoft.com/en-us/office/excel-specifications-and-limits-1672b34d-7043-467e-8e27-269d656771c3>

Business Process Automation with Python

Motivation – Extensibility

- Python in Excel ¹



1. *Announcing Python in Excel: Combining the power of Python and the flexibility of Excel.* TECHCOMMUNITY.MICROSOFT.COM.
<https://techcommunity.microsoft.com/t5/excel-blog/announcing-python-in-excel-combining-the-power-of-python-and-the/ba-p/3893439>

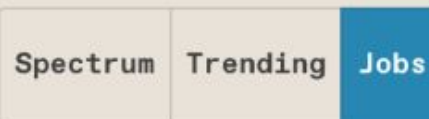
Business Process Automation with Python

Motivation – Jobs

- Python → 1 of the top languages for work

Top Programming Languages 2024

Click a button to see a differently weighted ranking



1. The Top Programming Languages 2024. (2024 August 22). IEEE Spectrum.
<https://spectrum.ieee.org/top-programming-languages-2024>

Business Process Automation with Python

Motivation – Jobs

- Python → Recently added as part of CFA exams

Level II

- ✓ **Python Programming Fundamentals**

A fundamentals course to demonstrate the basics of Python and how to use Jupyter Notebook for developing, presenting, and sharing data science projects related to finance. (if not taken at Level I)

- ✓ **Analyst Skills**

Focuses on the skills equity and credit analysts need using insights gained from hundreds of successful analysts.

- ✓ **Python, Data Science & AI**

Introduces candidates to machine learning, artificial intelligence, and data science to understand financial statements, reporting, and analysis using Python.

1. *Practical Skills Module | CFA Program Evolution.* Practical Skills Module | CFA Program Evolution.
<https://evolve.cfainstitute.org/practical-skills-modules.html>

Business Process Automation with Python

Python runtime

- The language and basic runtime environment
- Just like Apple iOS, it keeps updating (e.g., Python 3.12.2)
 - Python 2 was no longer supported from 2020¹

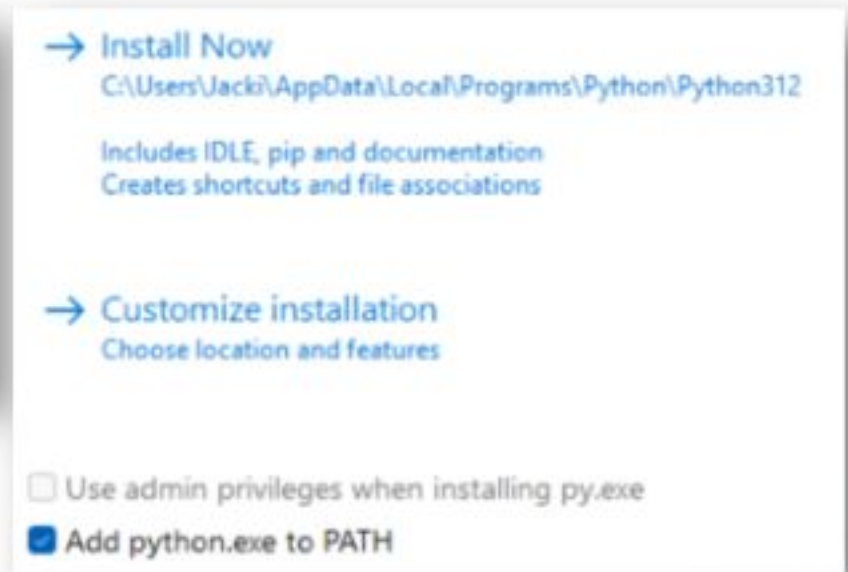


1. Sunsetting Python 2. Python.org. <https://www.python.org/doc/sunset-python-2/>

Business Process Automation with Python

Installation – Python Runtime

- Official website
 - <https://www.python.org/downloads/>
- Installing in Windows → Tick “Add python.exe to PATH”



Methology on how to run Python

Method 1 — Python Interpreter

- Install Anaconda Distribution in <https://www.anaconda.com/download>
- Open a terminal or command prompt.
- Type `python` and press Enter.
- Enter Python commands one by one and see the output immediately.

Method 2 — Colab

- Go to colab.research.google.com.
- Sign in with your Google account.
- Create a new notebook.
- Write and run Python code in cells using Shift+Enter.

Method 3 — Jupyter notebook

- Run `jupyter notebook` in terminal.
- Open the browser interface.
- Create a new notebook and run cells with Shift+Enter.

Method 4 — Script Mode

- Write code in a text editor and save it as a `.py` file.
- Run it from the terminal using `python filename.py`.
- Example: `python main.py`.

Python Interpreter

- Fast to start, with low resource use
- Small Experiments Quick feedback
- Better to test python library
- Use PC CPU/GPU Resources.

Colab

- Runs in the cloud -> no local setup
- Easy to share and collaborate
- Free GPU/TPU Resources
- Data store in cloud with lower security
- Works well with Google Drive

Jupyter notebook

- Better interactive exploration.
- Open the browser interface.
- Better Documentation along code
- Data store locally, great security.

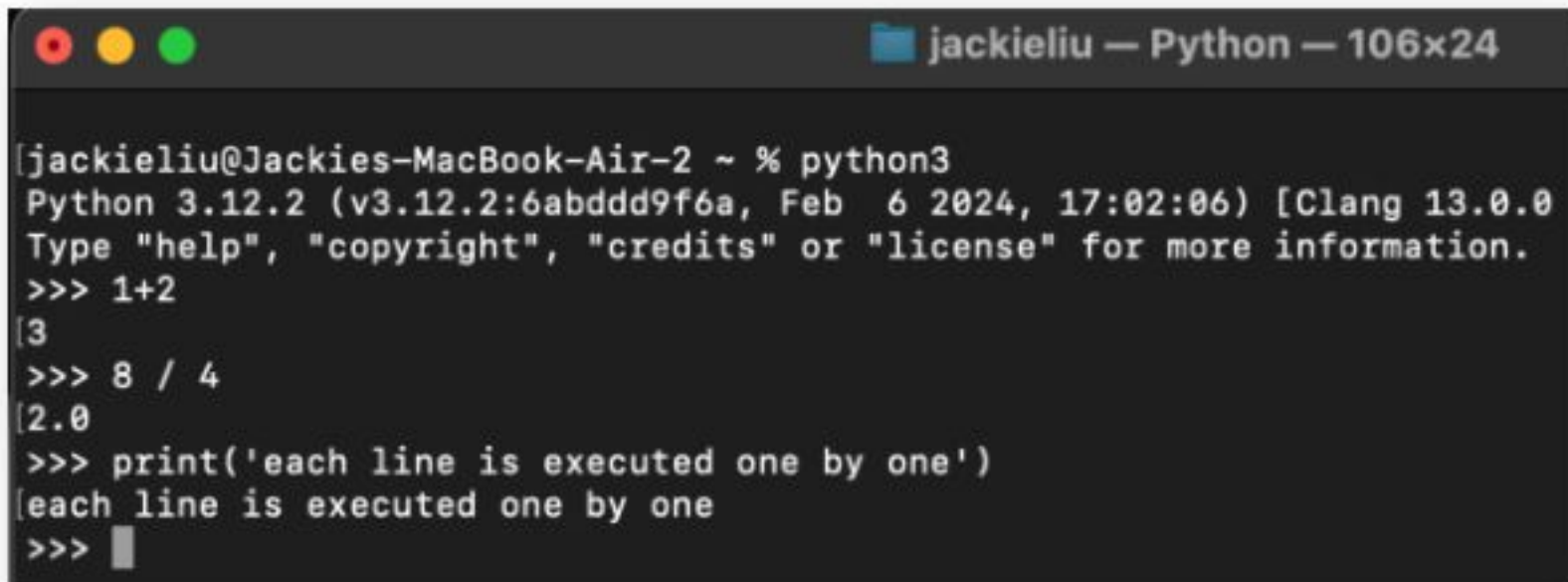
Method 4 — Script Mode

- Best for larger programs because code is saved in a structured file
- Easier to reuse, test, and deploy as a real application or module

Business Process Automation with Python

How to run Python – In the old days

- Python interpreter
 - This is the core of Python
 - Input a line of codes → Press Enter → Run

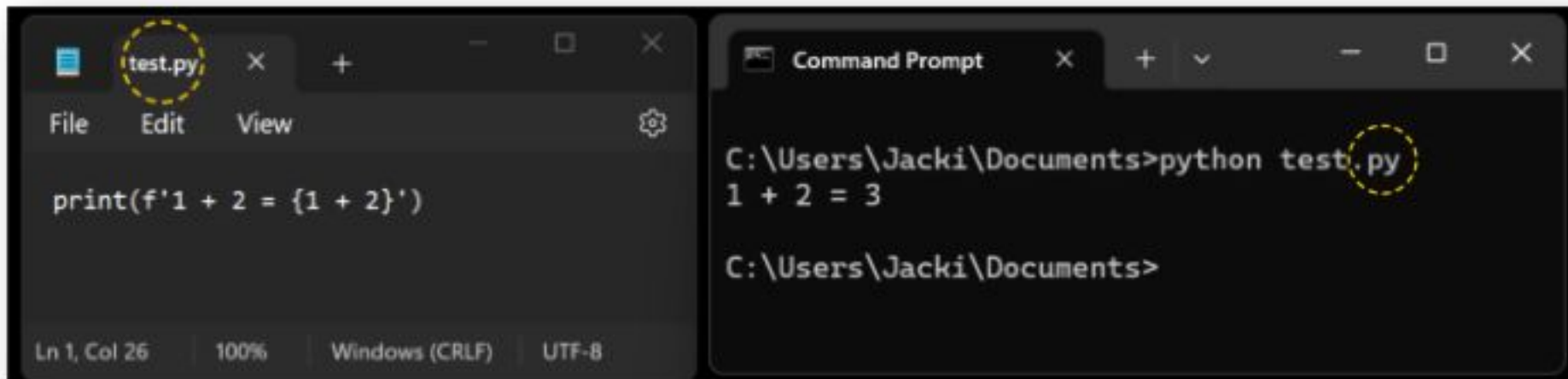


```
[jackieliu@Jackies-MacBook-Air-2 ~ % python3
Python 3.12.2 (v3.12.2:6abddd9f6a, Feb  6 2024, 17:02:06) [Clang 13.0.0
Type "help", "copyright", "credits" or "license" for more information.
>>> 1+2
[3
>>> 8 / 4
[2.0
>>> print('each line is executed one by one')
each line is executed one by one
>>> ]
```

Business Process Automation with Python

How to run Python – Running as a file

- Python interpreter + Text editor
 - Write codes in a text editor (e.g., Notepad, Vim, Sublime Text)
 - Save codes as a “.py” file
 - Run all the lines in one go

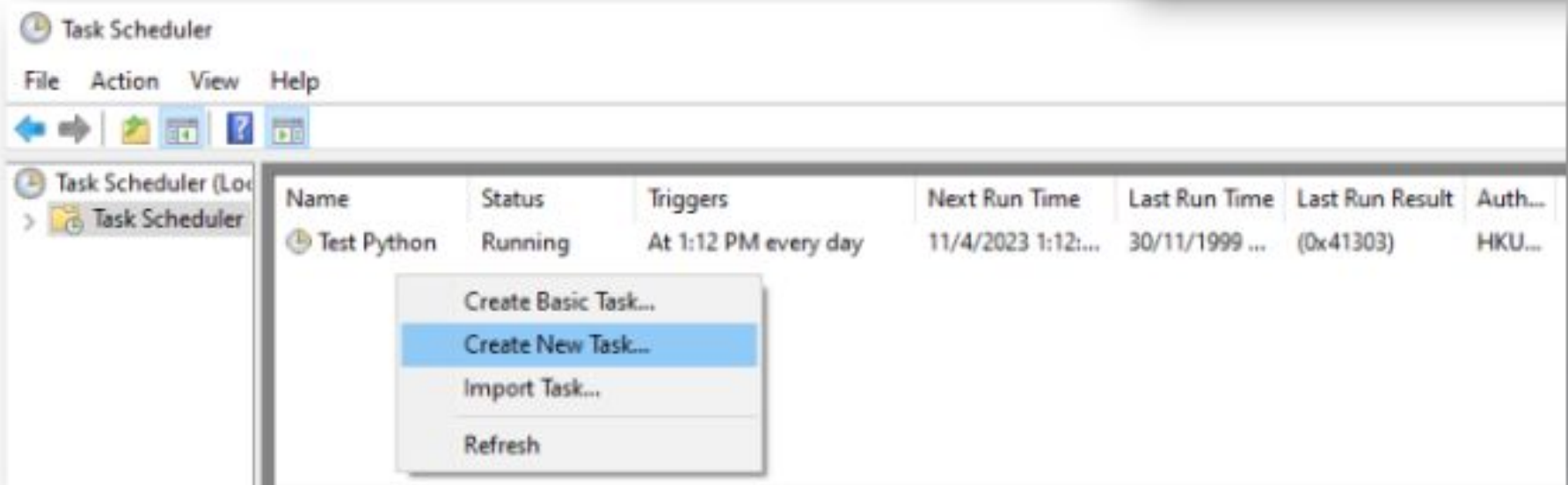
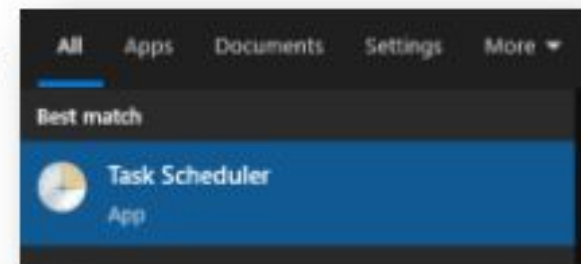


The screenshot shows two windows side-by-side. The left window is a text editor with a tab labeled 'test.py'. The code inside is `print(f'1 + 2 = {1 + 2}')`. The status bar at the bottom indicates 'Ln 1, Col 26', '100%', 'Windows (CRLF)', and 'UTF-8'. The right window is a 'Command Prompt' showing the command `C:\Users\Jacki\Documents>python test.py` being executed, followed by the output `1 + 2 = 3`. The prompt then returns to `C:\Users\Jacki\Documents>`.

Business Process Automation with Python

How to run Python – Running as a file (bonus)

- Python interpreter + Text editor + **Task Scheduler**
 - Running a Python script daily/ hourly
 - e.g., Task Scheduler/ Cron / Autosys etc
 - Create a task pointing to the .py file



Business Process Automation with Python

How to run Python – Summary

- Python interpreter
- Python (.py) files with
 - Text editors
 - IDEs
- Jupyter Notebook running
 - Locally
 - Remotely



Business Process Automation with Python

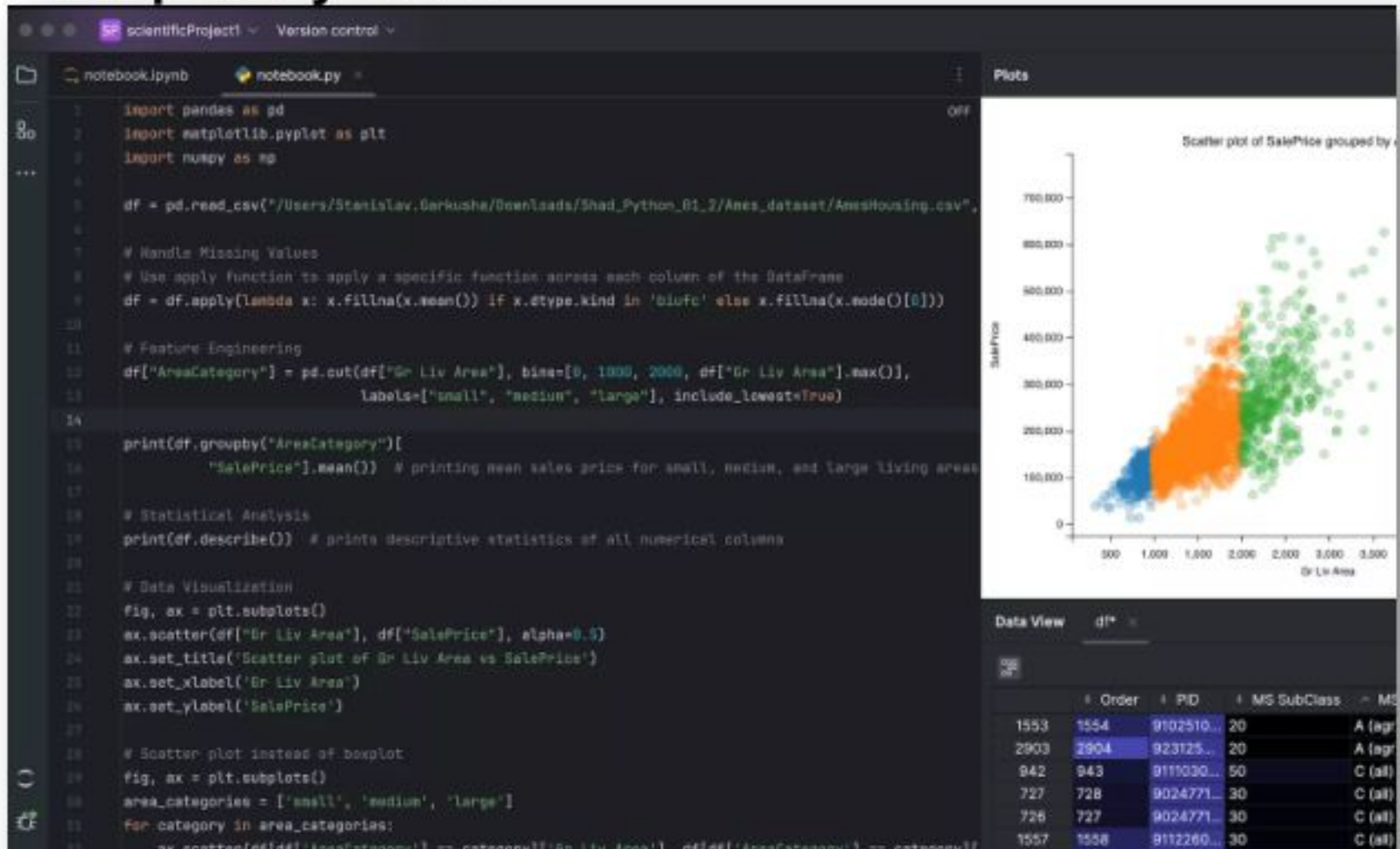
How to run Python – IDE

- Integrated Development Environment (IDE)
 - A smart editor tailored for writing codes
 - e.g., Syntax highlighting, variable tracking, debugging mode
 - May support useful extensions/plugins (e.g., GitHub Copilot)
 - Common IDEs
 - e.g., Spyder, PyCharm, Visual Studio Code



Business Process Automation with Python

Example - PyCharm



How to run Python – Jupyter Notebook

- Google Colaboratory (Colab)
 - Jupyter Notebook powered by Google Cloud ¹
 - <https://colab.research.google.com/>
 - Benefits ²
 - “Zero configuration required”
 - Easy sharing
 - Free access: CPU & GPU



1. Google Colab FAQ. <https://research.google.com/colaboratory/faq.html>

2. Google Colaboratory. <https://colab.research.google.com>

Cloud Software Introduction

Colaboratory, or Colab for short, is a Google Research product, which allows developers to write and execute Python code through their browser. Google Colab is an excellent tool for deep learning tasks. It is a hosted Jupyter notebook that requires no setup and has an excellent free version, which gives free access to Google computing resources such as GPUs and TPUs.

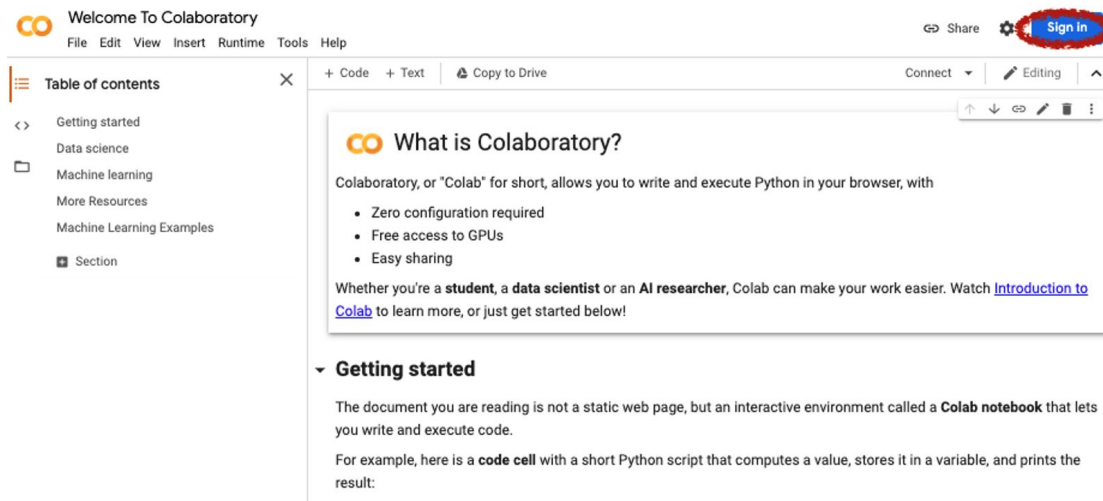
Why Should I Use Google Colab?

There are several reasons to opt to use Google Colab instead of a plain Jupyter Notebook instance:

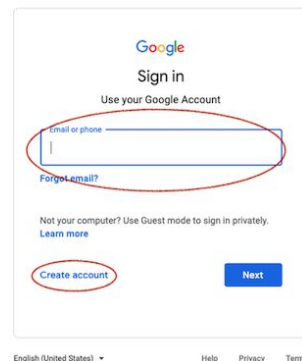
- Pre-Installed Libraries
- Saved on the Cloud
- Collaboration
- Free GPU and TPU Use

Google Colab Setup

1. Visit the [Google Colab](http://colab.research.google.com/) (<http://colab.research.google.com/>) page, which will direct you to the [Google Colaboratory Welcome Page](#).
2. Click the **Sign in** button on the right top.

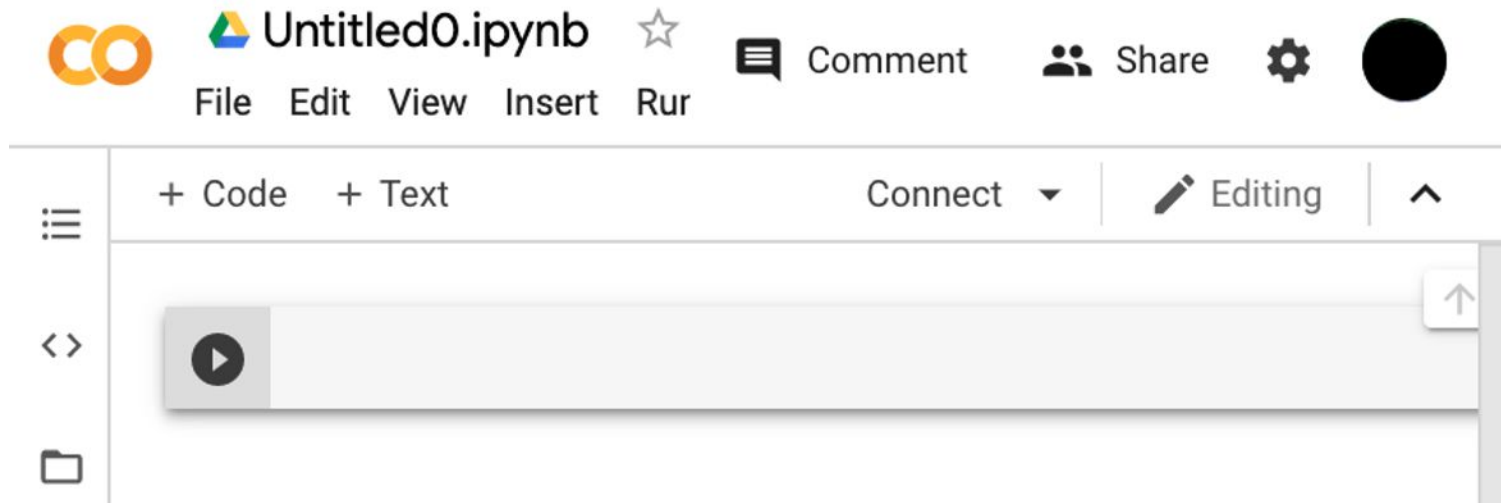


1. Sign in with your Gmail account.



Google Colab Setup

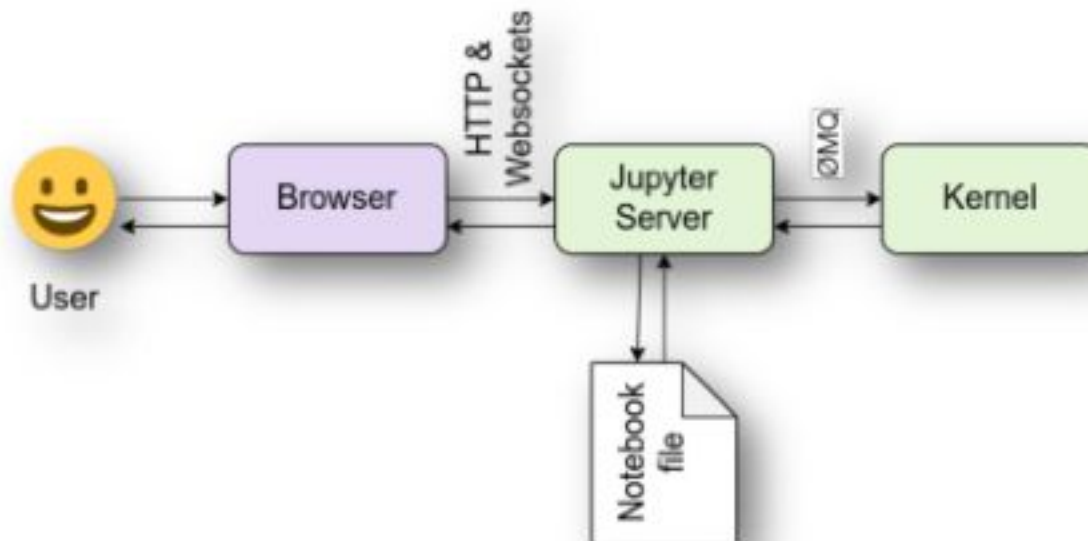
4. As soon as you complete the sign-in process, you are ready to use Google Colab.
4. You may easily create a new Colab notebook on this page by clicking *File> New notebook*.



Business Process Automation with Python

How to run Python – Jupyter Notebook

- Web-based interactive platform
 - Accessible: Python runtime on a web server

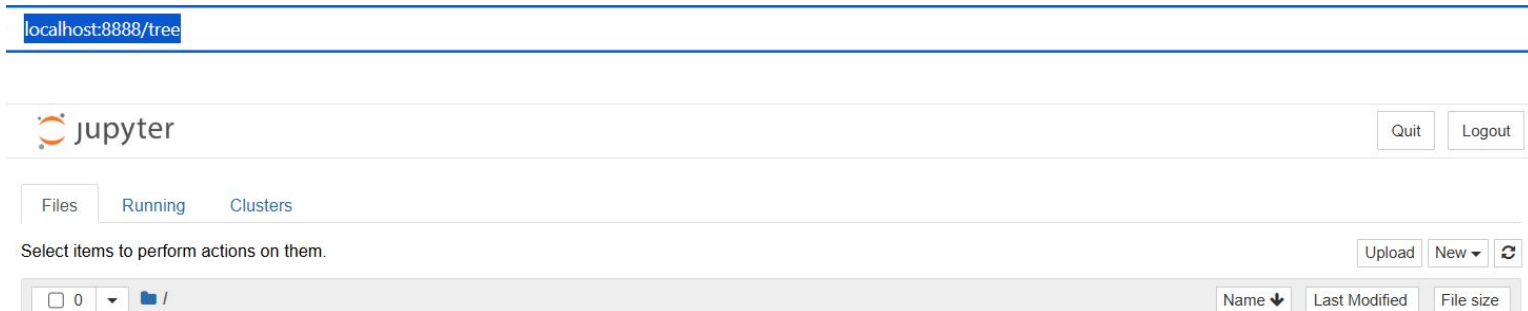


1. Architecture — Jupyter Documentation 4.1.1 Alpha Documentation.
<https://docs.jupyter.org/en/latest/projects/architecture/content-architecture.html>

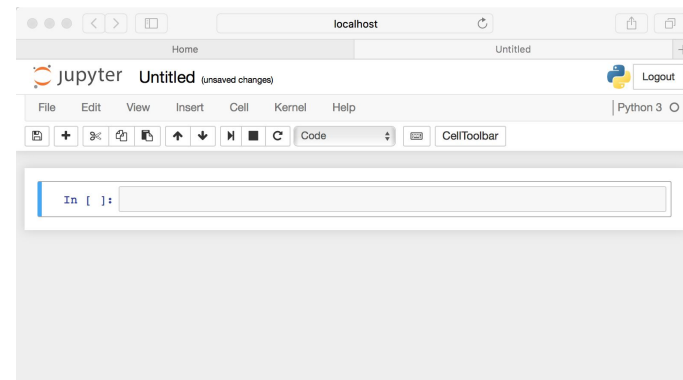
Python Coding Environment Practise - Jupyter notebook

Open Jupyter notebook

1. In Desktop, click 'Startup' button
2. Type "Anaconda Prompt (Anaconda3)"
3. Type "jupyter notebook" to trigger your first notebook
4. <http://localhost:8888/tree> is opened



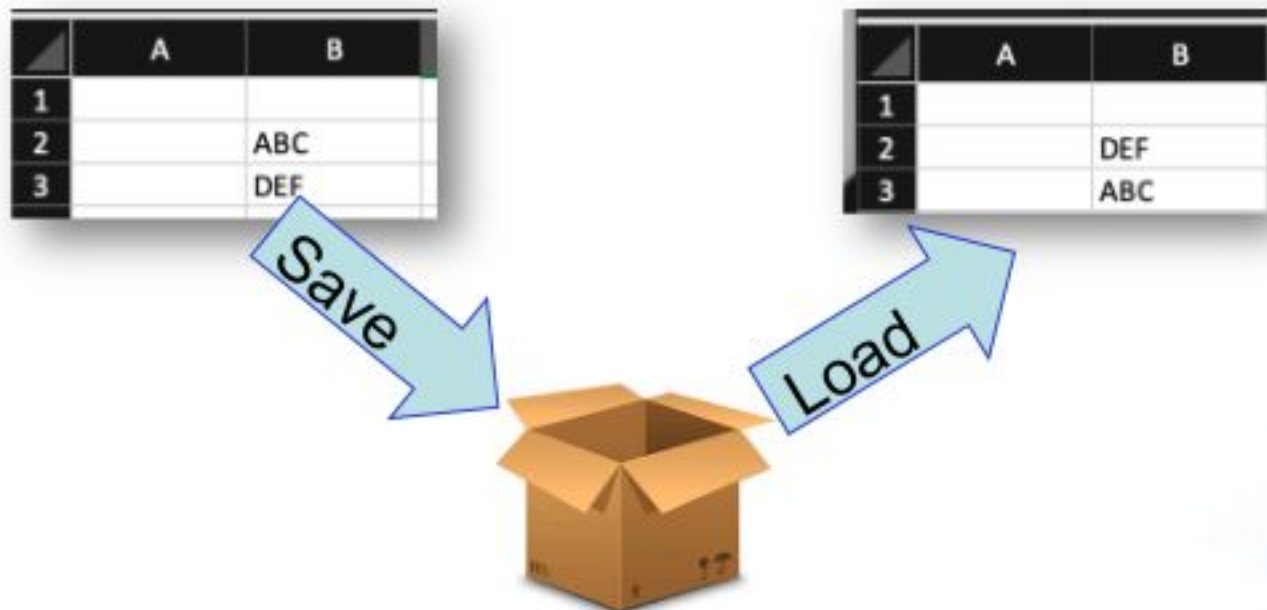
1. In right top side, click New -> Python 3



Business Process Automation with Python

Variables

- From the previous lecture
 - Variables → Storage of **values**
 - **Values** → carry data types



Business Process Automation with Python

Variables

- Declaration
 - In VBA, declarations (“dim”) are recommended
 - Dim year As **Integer**
 - year = 2024
 - In Python, variables are created during **assignment**
 - year = 2024
 - To assign without an actual value
 - year = None



Business Process Automation with Python

Value Check

- VBA
 - Debug.Print()
- Python
 - print()
 - Last line in a block

```
Sub HighlightCell()  
    If Range("B1").Value > 50 Then  
        Range("B1").Interior.Color = vbYellow  
    Else  
        Range("B1").Interior.Color = vbGreen  
    End If  
    Debug.Print (Range("B1"))  
End Sub
```

```
▶ print(1 + 2)           # This gives 3  
  print(1 + 2 + 3)       # This gives 6
```

3
6

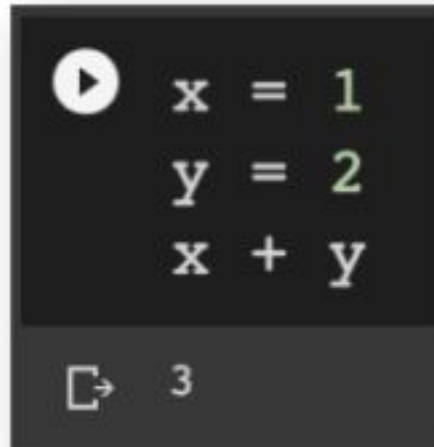
```
▶ 1 + 2                 # This is not shown  
  1 + 2 + 3             # This is shown
```

6

Business Process Automation with Python

Primitive Data Types - Numeric

- Int

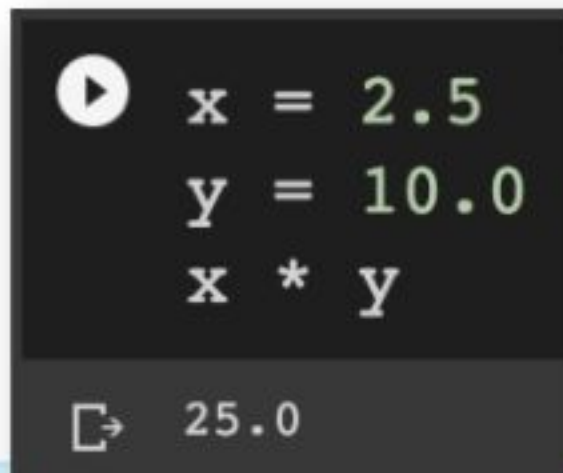


```
x = 1
y = 2
x + y
```

3

A dark-themed code editor snippet showing three lines of Python code: `x = 1`, `y = 2`, and `x + y`. The code is executed, and the result `3` is displayed at the bottom.

- Float



```
x = 2.5
y = 10.0
x * y
```

25.0

A dark-themed code editor snippet showing three lines of Python code: `x = 2.5`, `y = 10.0`, and `x * y`. The code is executed, and the result `25.0` is displayed at the bottom.

Primitive Data Types - Numeric

- Complex (For scientific calculations)

THE QUADRATIC FORMULA

© CHILMATH.COM

If $ax^2 + bx + c = 0$ but $a \neq 0$

then

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

DISCRIMINANT

- $b^2 - 4ac > 0$ two real solutions
- $b^2 - 4ac = 0$ one real solutions
- $b^2 - 4ac < 0$ zero real solutions

$$\begin{aligned}x &= 3 + 4j \\y &= 2 + 2j \\x - y\end{aligned}$$

$$\Rightarrow (1+2j)$$

$$i = \sqrt{-1}$$

Business Process Automation with Python

Primitive Data Types - Numeric

- Common Functions
 - `round()` / `pow()` / `abs()`

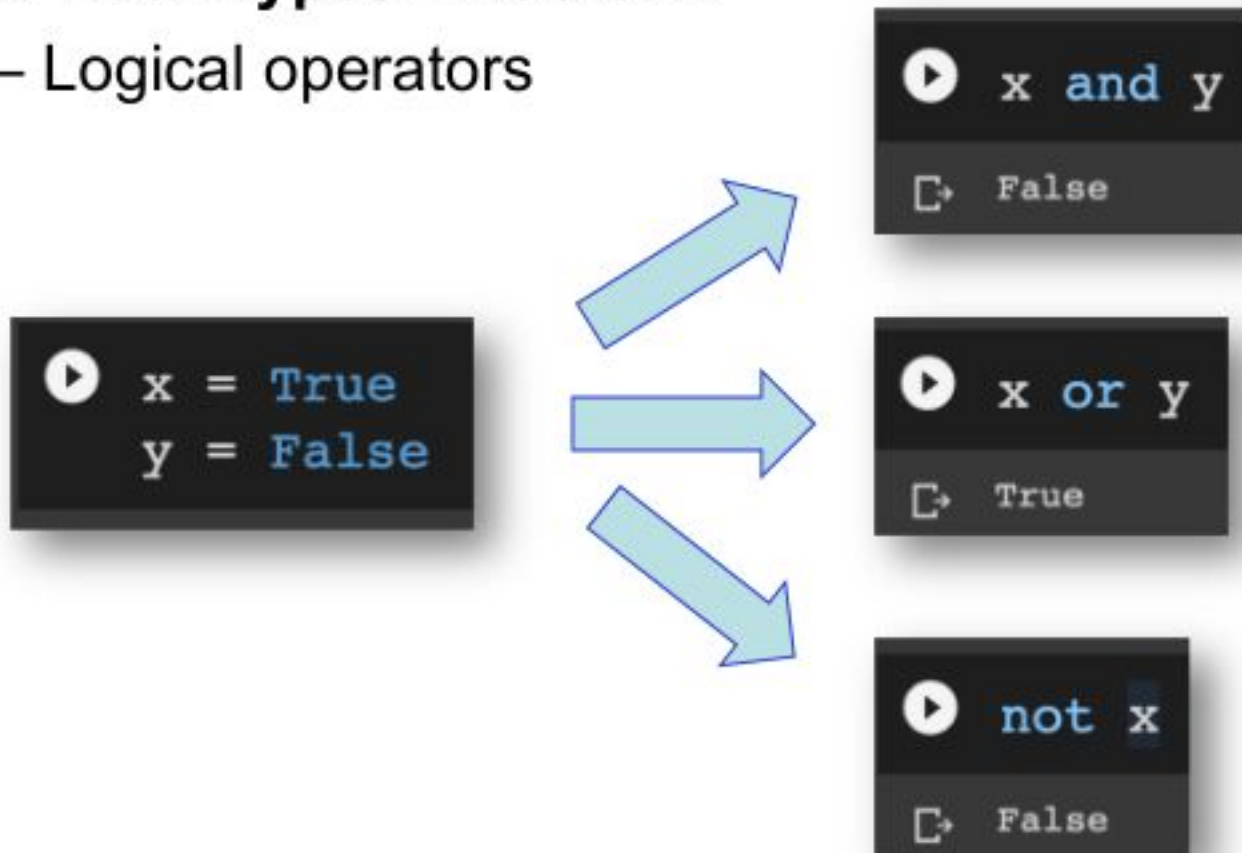
```
▶ print(round(9876.54321, 2))    # Round to 2 decimal places
  print(round(9876.54321, 0))    # Round to nearest integer
  print(pow(2, 3))               # 2 * 2 * 2 = 8
  print(abs(-5))                # Negative numbers become positive
```

```
□ 9876.54
   9877.0
   8
   5
```

Business Process Automation with Python

Primitive Data Types - Boolean

- Bool – Logical operators



Primitive Data Types - Boolean

- Bool – comparators
 - Generates True/ False

Operator	Name	Example
==	Equal	5 == 5
!=	Not equal	26 != 3
>	Greater than	100 > 67
<	Less than	89 < 216
>=	Greater than or equal to	90 >= 54
<=	Less than or equal to	23 <= 77

Business Process Automation with Python

Primitive Data Types - Text

- String
 - Representation

```
▶ x = 'This is a string'           # Single quote
  y = "This is also a string"      # Double quote
  z = '''This is a very long string
spanning across more than 1 line''' # Multi-line

print(x)
print(y)
print(z)
```

```
☞ This is a string
   This is also a string
   This is a very long string
   spanning across more than 1 line
```

Business Process Automation with Python

Primitive Data Types - Text

- String
 - Concatenation → "+" → glue 2 strings together

```
▶ x = 'This is a string'  
  y = "This is also a string"
```

```
▶ print(x + ' and ' + y) # Adding 2 strings together
```

```
☞ This is a string and This is also a string
```

Business Process Automation with Python

Primitive Data Types - Text

- String
 - Concatenation → Number vs Text



```
lobster_price = 936
lobster_text = str(lobster_price)
print(2 * lobster_price) # 1872
print(2 * lobster_text) # 936936
```

1872
936936

```
print(10 * '=')
print('WELCOME')
print(10 * '=')
```

```
=====
WELCOME
=====
```

1. 星島日報. (2023, May 6). 荳蔻中菜館驚現「天價龍蝦」兩隻竟索價90萬元 酒樓親解釋. Singtaousa.com; 星島日報.
<https://www.singtaousa.com/2023-05-06/%e8%8d%83%e7%81%a3%e4%b8%ad%e8%8f%9c%e9%a4%a8%e9%a9%9a%e7%8f%be%e3%80%8c%e5%a4%a9%e5%83%b9%e9%be%8d%e8%9d%a6%e3%80%8d-%e5%85%a9%e9%9a%bb%e7%ab%9f%e7%b4%a2%a5%83%b990%e8%90%ac%e5%85%83-%e9%85%92/4488305>

Business Process Automation with Python

Primitive Data Types - String formatting

- F-String (Python 3.6 or newer)
 - A smart way to format strings

```
▶ x = 4  
  y = 8.8  
  f'x is {x} and y is {y}; y divided by x is {y / x}'  
□→ 'x is 4 and y is 8.8; y divided by x is 2.2'
```

1. *Input and Output*. Python Documentation. <https://docs.python.org/3/tutorial/inputoutput.html>

Business Process Automation with Python

Primitive Data Types - String formatting

- F-String (Python 3.6 or newer)
 - A smart way to format strings (advanced)

```
▶ z = 1234.56789  
print(z)  
print(f'{z:.2f}')print(f'{z:,.2f}')
```

```
☞ 1234.56789  
1234.57  
1,234.57
```

1. *Input and Output*. Python Documentation. <https://docs.python.org/3/tutorial/inputoutput.html>

Primitive Data Types - String

- Common Functions
 - upper() / lower() / replace()

```
▶ # Common functions  
test_string = 'Hello Hong Kong!'  
print(test_string.upper())  
print(test_string.lower())  
print(test_string.replace('Hello', 'Bello'))
```

```
☞ HELLO HONG KONG!  
hello hong kong!  
Bello Hong Kong!
```

Business Process Automation with Python

Primitive Data Types - Conversion

- **VBA**

- CInt() / CDbl() / Format()

- **Python**

- int() / float() / str() / bool()

 `int(4.5)` 4 `float(5)` 5.0 `str(5.55)` '5.55' `bool('Some values')` True `bool('')` False

Business Process Automation with Python

Practice 1 – Market Capitalization

- Compute market capitalization of 2 companies
- Show the difference between the two
- Format the result up to 2 decimal places


$$\text{Market Capitalization Formula} = \text{Current Market Price Per Share} \times \text{Total Number of Outstanding Shares}$$


- **Starting notebook:**

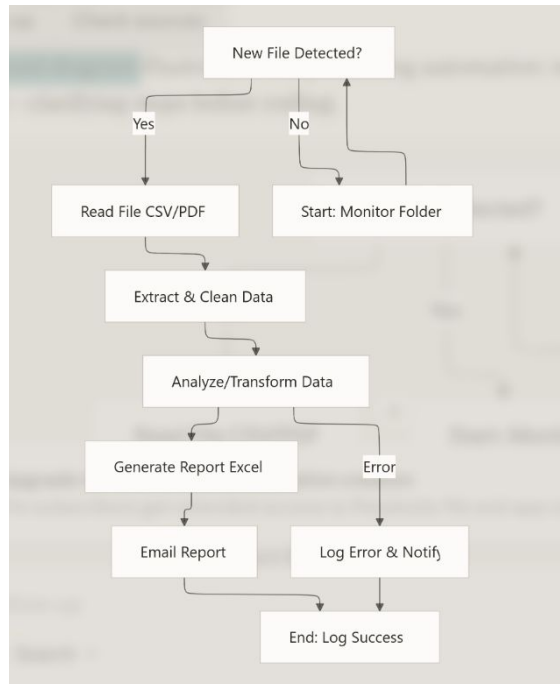
https://github.com/innoviai/IPA_Courses-May2026/blob/main/practice1_simple_calc_202409.ipynb

Python Process Automation Overview

Python streamlines working processes by automating tasks like data extraction and reporting, minimizing manual effort.

The objective of drawing the flowchart is to visually map the automation sequence, aiding in design, debugging, and stakeholder communication for reliable implementation.

Methodology of Workflow Flowchart: Mermaid Diagram



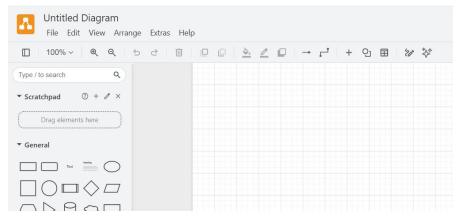
Software to draw Flowchart : draw.io

Key Benefits

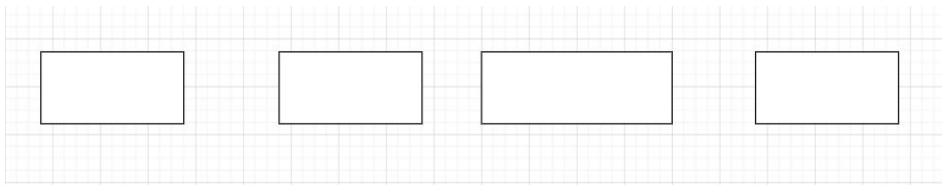
- draw.io (now diagrams.net) is a free, web-based diagramming tool ideal for creating flowcharts, offering drag-and-drop shapes, templates.

Quick Steps to Create a Flowchart

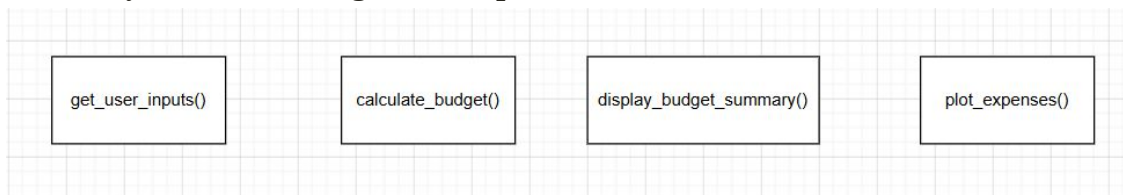
- Go to app.diagrams.net.



- Conduct an python flow chart with title “How to Manage Your Budget”
- Drag 4 shapes “Rectangle” from left panel into main canvas

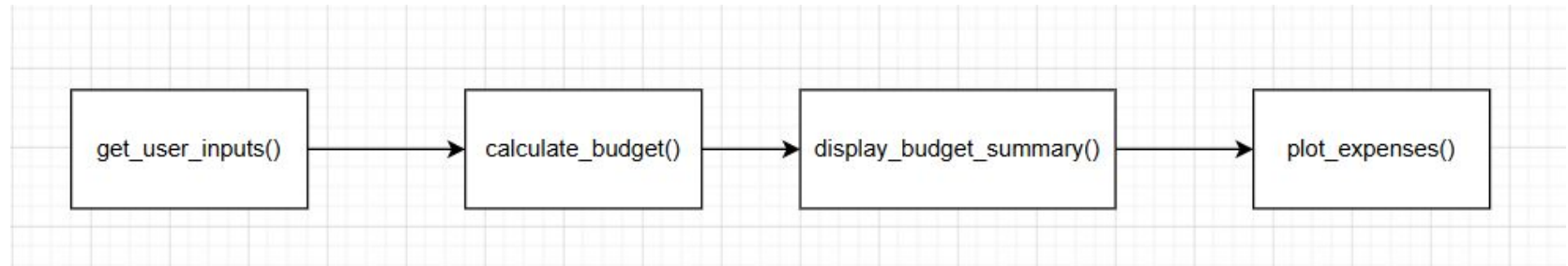


- Add labels by double-clicking each shapes.

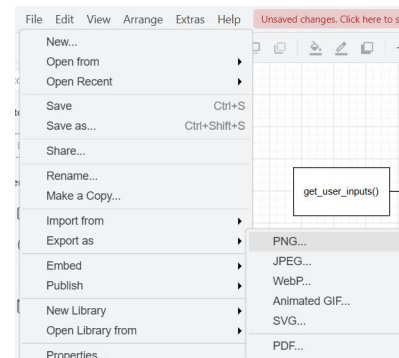


Software to draw Flow Charts: draw.io

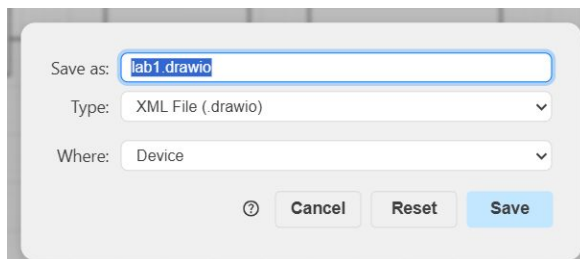
5. Connect with arrows



6. Click "File" -> "Export As" and select either "PNG/PDF"



7. or Click "Save As" and assign an file name "lab1.drawio" and save to Device/Google Drive/OneDrive..



Business Process Automation with Python

Methods

- VBA
 - Sub-routine: reusable code blocks
 - Function: sub-routine which gives an output
- Python
 - Methods: Use “return” if output is required

VBA

```
Function AddNumbers(x As Double, y As Double) As Double  
    AddNumbers = x + y  
End Function
```

?AddNumbers(1,2)
3

Python

```
def add_numbers(x, y):  
    return x + y
```

```
def add_numbers(x, y):  
    return x + y  
  
print(add_numbers(1,2))
```


Open Jupyter notebook

1. Getting User Input

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
def get_user_input():
```

```
    """Get user input for income and expenses."""
```

```
    income = float(input("Enter your income: "))
```

```
    # Expecting expenses to be a dictionary input
```

```
    expenses = {}
```

```
    while True:
```

```
        category = input("Enter expense category (or 'done' to finish): ")
```

```
        if category.lower() == 'done':
```

```
            break
```

```
        amount = float(input(f"Enter amount for {category}: "))
```

```
        expenses[category] = amount
```

```
    return income, expenses
```

How to Manage Your Budget with a Simple Python Script

2. Calculating the Budget

```
def calculate_budget(income, expenses):  
    """Calculate total expenses and balance."""  
    total_expenses = sum(expenses.values())  
    balance = income - total_expenses  
    return total_expenses, balance
```

3. Displaying the Budget Summary

```
def display_budget_summary(income, total_expenses, balance):  
    """Display the budget summary."""  
    print(f"Income: ${income:.2f}")  
    print(f"Total Expenses: ${total_expenses:.2f}")  
    print(f"Balance: ${balance:.2f}")
```

How to Manage Your Budget with a Simple Python Script

4. Plotting the Expenses

```
def plot_expenses(expenses):  
    """Plot the expenses as a bar chart."""  
    df = pd.DataFrame(list(expenses.items()), columns=['Category', 'Amount'])  
    df.plot(kind='bar', x='Category', y='Amount', legend=False)  
    plt.ylabel('Amount ($)')  
    plt.title('Expense Distribution')  
    plt.show()
```

How to Manage Your Budget with a Simple Python Script

5. Main Function

```
income, expenses = get_user_input()
total_expenses, balance = calculate_budget(income, expenses)
display_budget_summary(income, total_expenses, balance)
plot_expenses(expenses)
```

Core Concepts

Each line represents a sequential execution: gather input, compute results, display output, and visualize data. Variables like `income` and `expenses` hold **values**, passed as **arguments** to functions for processing.

Values Examples

- `income = 5000` (an integer value).
- `expenses = [1000, 2000, 500]` (a list value).
- `balance = 1500` (computed result value).

Arguments Examples

- `income` is the first argument.
- `expenses` is the second argument.
- `balance` is the third argument.

How to Manage Your Budget with a Simple Python Script

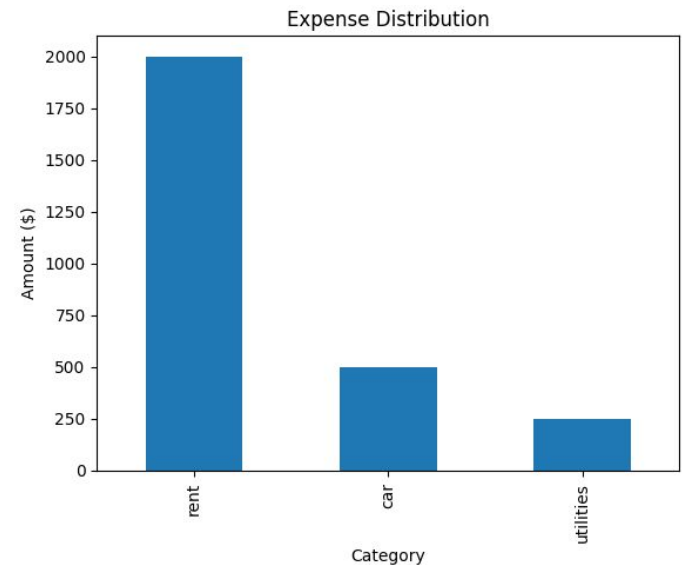
Run above Code and following is displayed

Main Input

Enter your income: 10000
Enter expense category (or 'done' to finish): rent
Enter amount for rent: 2000
Enter expense category (or 'done' to finish): car
Enter amount for car: 500
Enter expense category (or 'done' to finish): utilities
Enter amount for utilities: 250
Enter expense category (or 'done' to finish): done

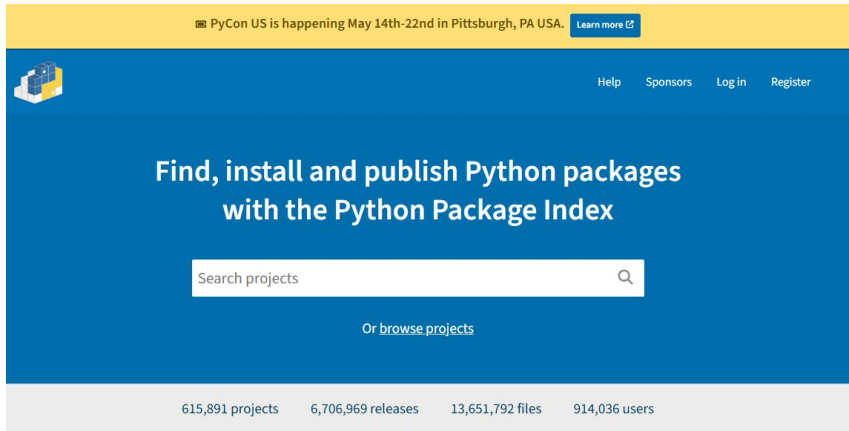
Result

Income: \$10000.00
Total Expenses: \$2750.00
Balance: \$7250.00

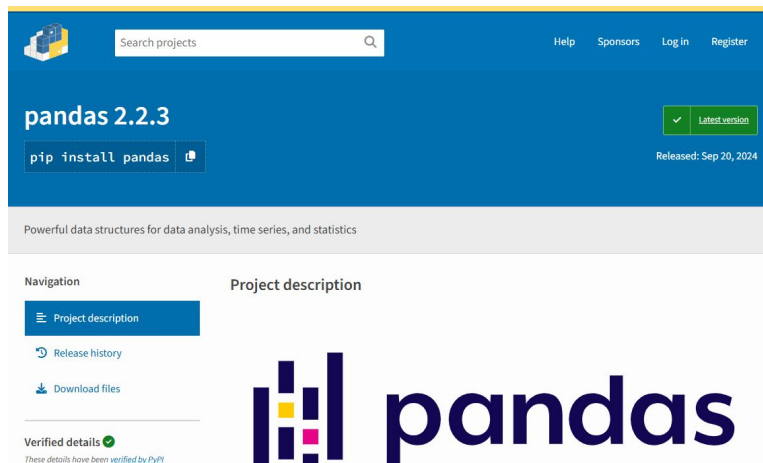


Library Installation in Python

Most of the python library can be founded in <https://pypi.org/>

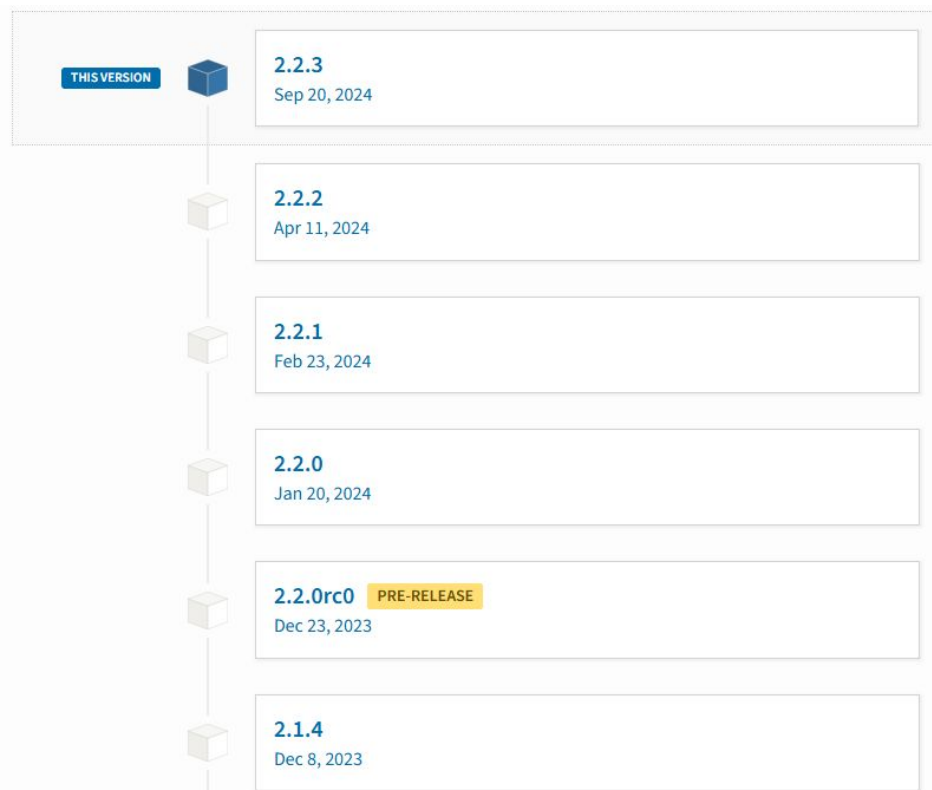


You can enter pandas to search or type <https://pypi.org/project/pandas/>



Library Installation in Python

Same library name have different version like 2.2.3, 2.1.0, ...
The default page show the newest version.



If you use python 3.7 or 3.8, you better install the version 2.1.0

pandas 2.1.0

```
pip install pandas==2.1.0
```



For library installation, you can go to anaconda prompt and enter
“pip install pandas” or “pip install pandas==2.1.0”

Introduction to Business Process Automation

Types of Business Automation – Advanced Automation

- **Target** → Complex process: Multiple systems + Human
- **Why**
 - Handles unstructured data / methods of access
- **Example** → Machine learning, natural language processing

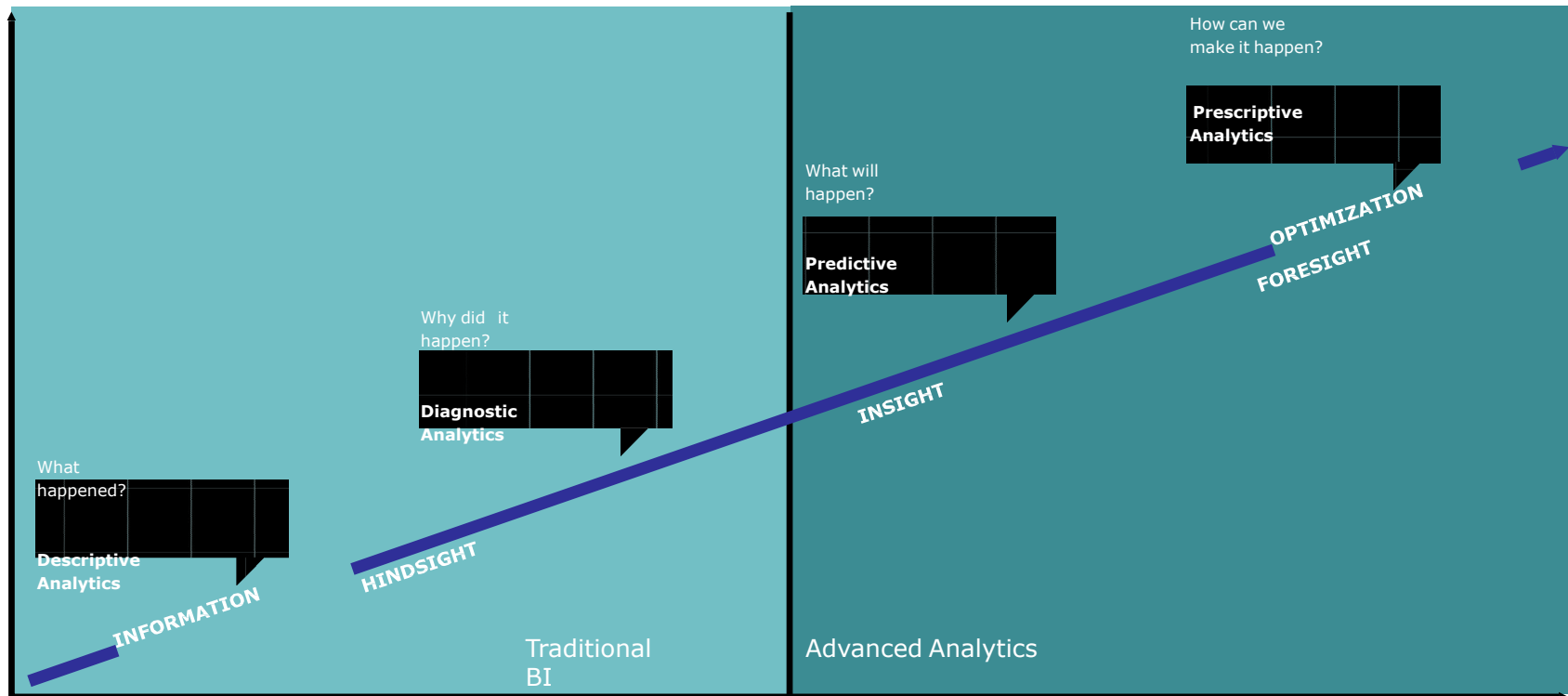


What Is Data Science?

From Wikipedia:

Data science, also known as data-driven science, is an interdisciplinary field about scientific processes and systems to extract knowledge or insights from data in various forms, either structured or unstructured, which is a continuation of some of the data analysis fields such as statistics, machine learning, data mining, and predictive analytics similar to Knowledge Discovery in Databases (KDD).

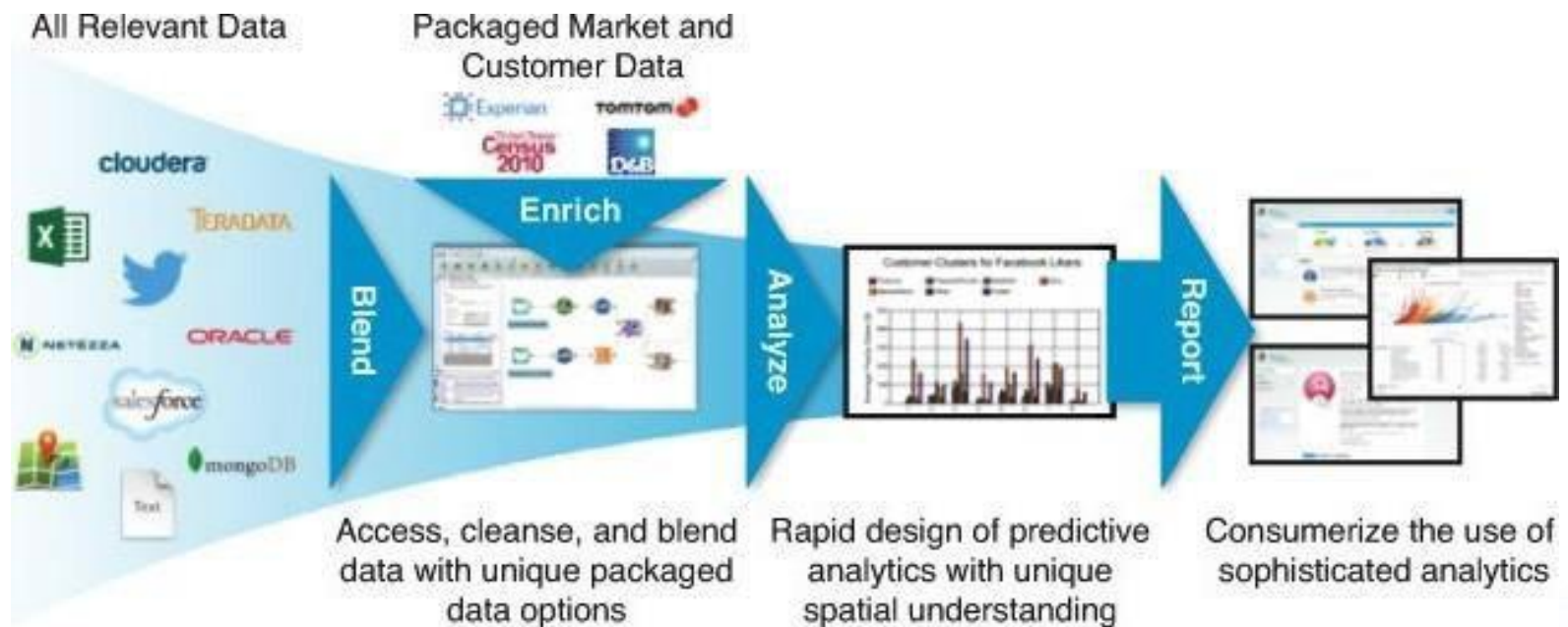
How Is Business Intelligence Different From Data Science?



Modern BI Process - Blend, Enrich, Analyze, and Report

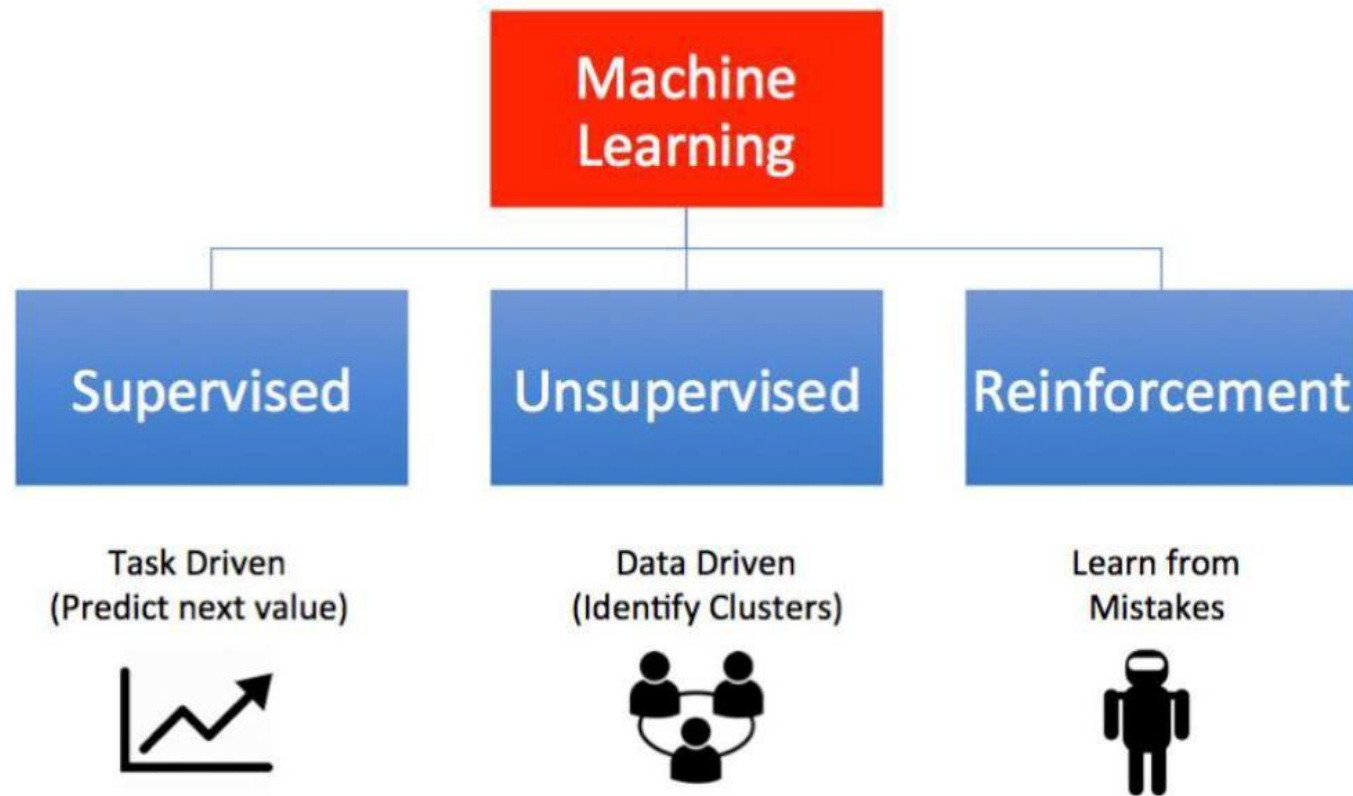
Predictive Analytics: By applying machine learning algorithms to historical project data, organizations can forecast outcomes more accurately and mitigate risks in project delivery

Business Intelligence: Automating the preparation of data for analytics tools allows organizations to gain actionable insights more rapidly, enhancing their decision-making capabilities.

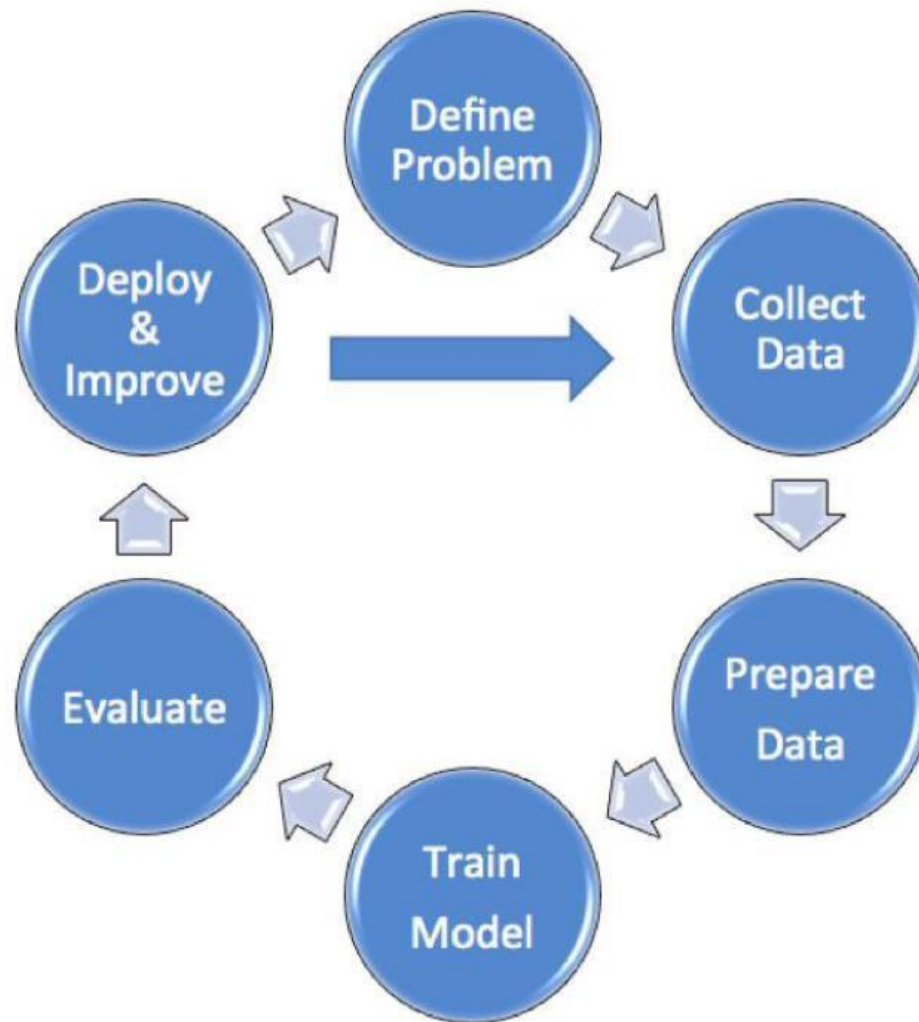


What is Business Analytics & Machine Learning?

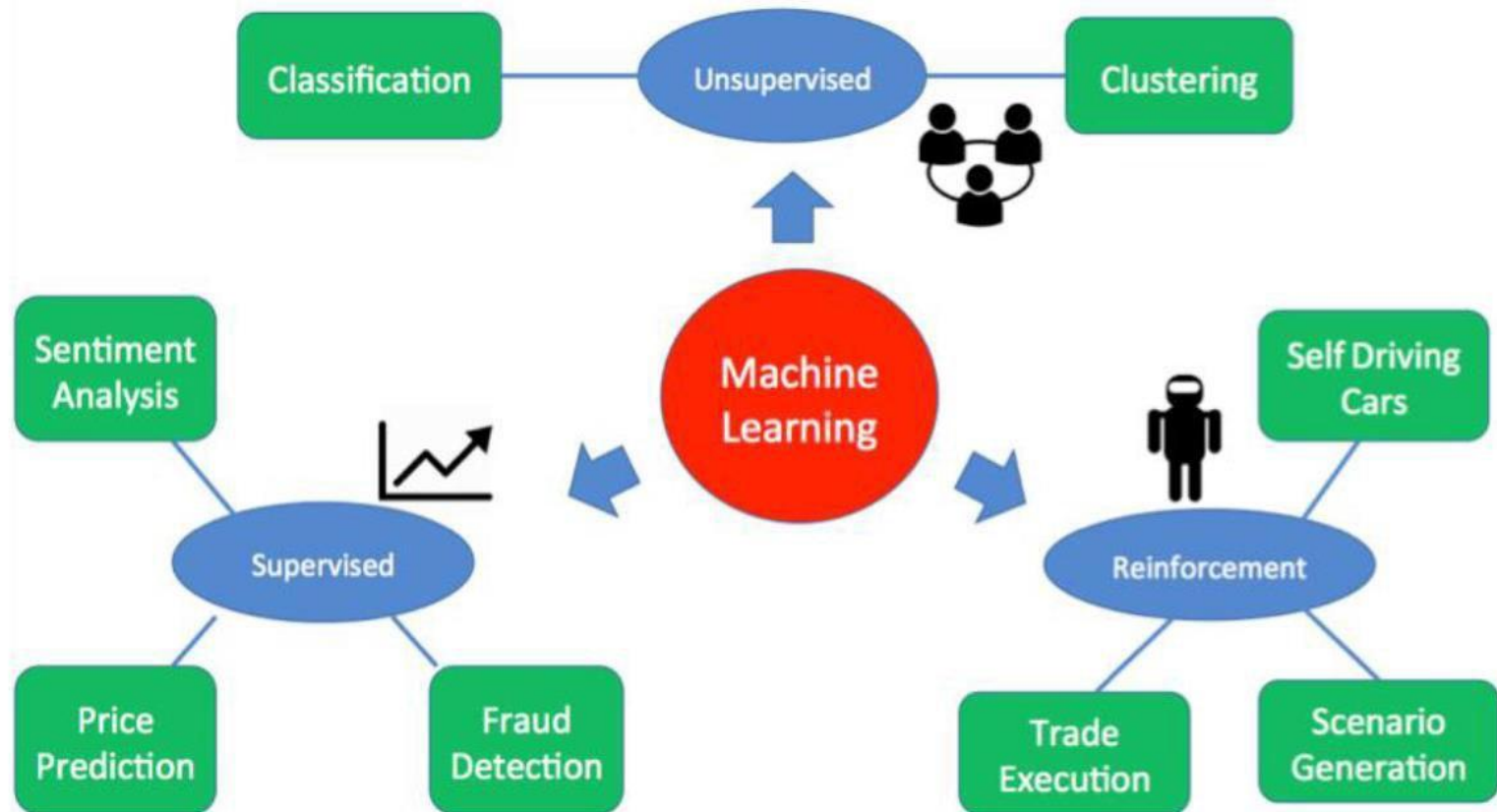
Types of Machine Learning



What is Business Analytics & Machine Learning?

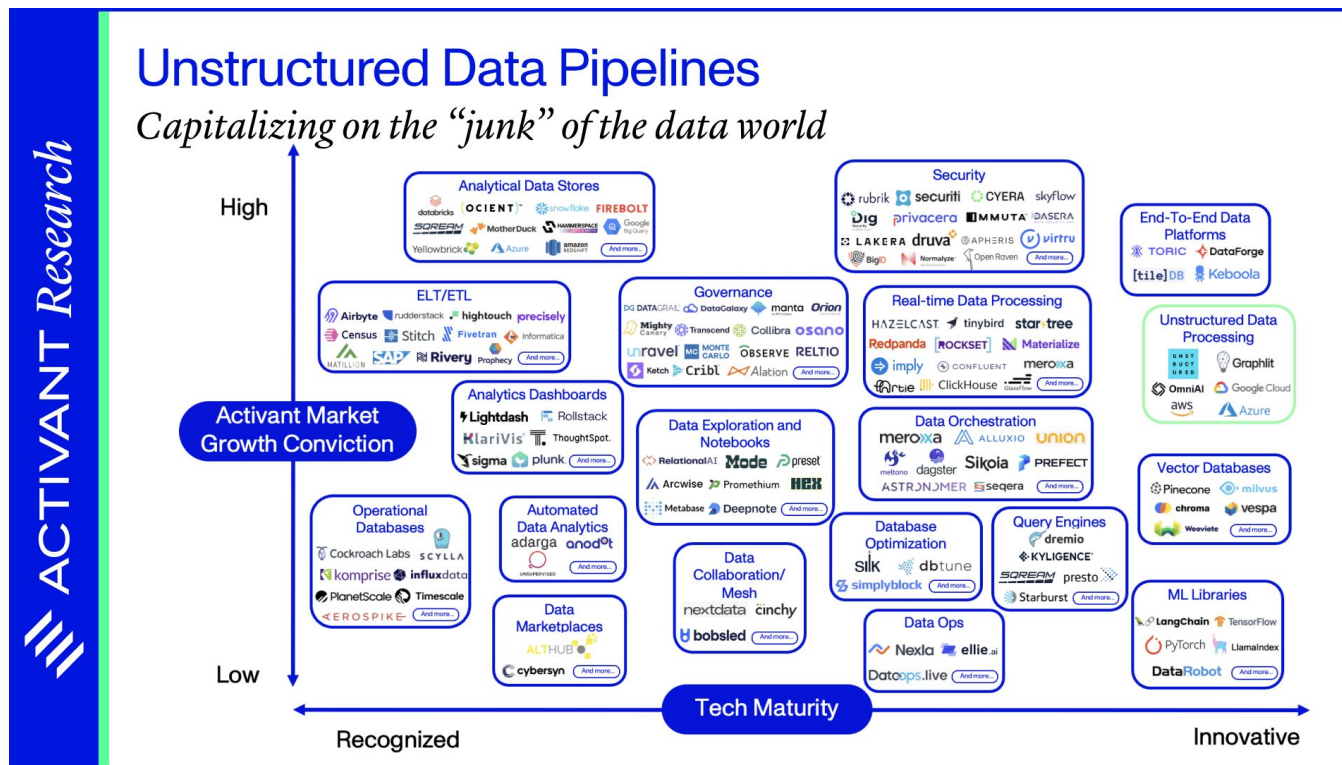


What is Business Analytics & Machine Learning?



IPA – Data Science

IPA's integration with data science is particularly impactful in several areas:
Unstructured Data Processing: IPA can convert unstructured data into structured formats suitable for analysis, enabling businesses to leverage insights from previously untapped data source.



Concept of Structured, Unstructured, Discrete, Continuous Data

Structured Data

- Fixed schema (rows/columns, e.g., SQL tables)
- Easy to query/search (names, prices)
- needs traditional ML models such as XGBoost, Random Forests techniques

Unstructured Data

- No fixed format (text, images, videos)
- needs deep learning vs NLP/ML techniques

Discrete Data

- Countable, distinct values (e.g., 0,1,2 items)
- needs traditional ML classification models like Logistic Regression or Decision Trees

Continuous Data

- Any value in range (e.g., 1.75m height)
- No gaps; line/histograms
- needs traditional ML regression models such as Linear Regression or Gradient Boosting

Data Processing & Analysis

Practise 2

**Which Data Type is Structured/Unstructured,
Discret/Continuous?**

E-mails	
Digital Images	
Stock Market Logs	
Historical Gold Prices	
Credit Approval Records	
Social Media Friend Relationships	
Tweets & Trending Topics	
Sales Records	

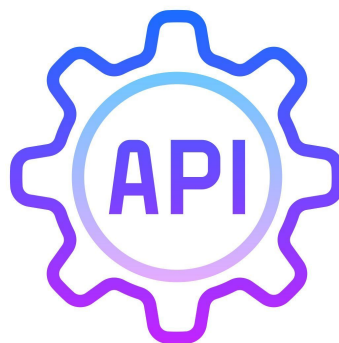
Source: Practical Data Analysis - Second Edition Hector Cuesta, Dr. Sampath Kumar OACKT Publishing BIRMINGHAM - MUMBAI

Structured Data Collection Method

Main Data collection are get from Google Sheets, API and Web Scrapping



Google Sheets



kaggle

<https://www.kaggle.com/>



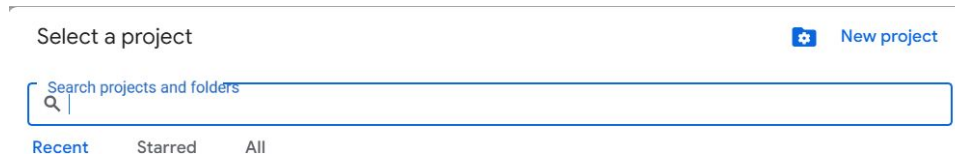
開放數據平台
DATA.GOV.HK

<https://data.gov.hk/en/>

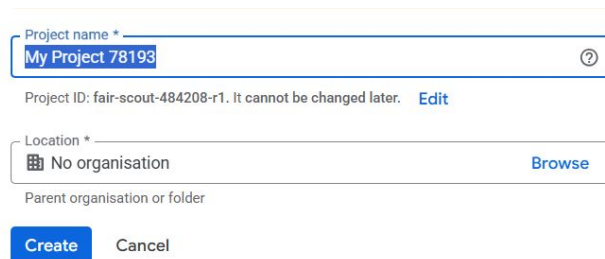
Data Collection - Reading Data from Google Sheet

Enable the Google Sheets API with readpermission.

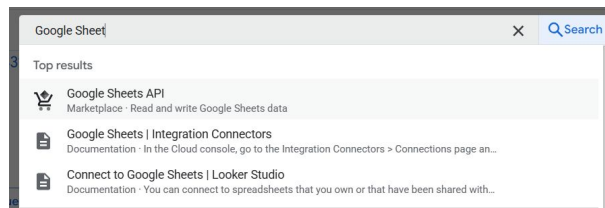
1. Open Browser and enter <https://console.cloud.google.com/>
2. Click “Select a project”
3. Click “New Project”



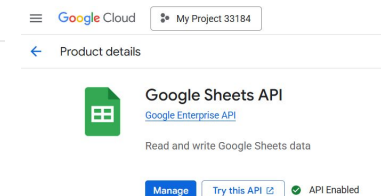
4. Enter information and click “Create”



5. Select your Project that create
6. in Search Bar, enter ‘Google Sheet’



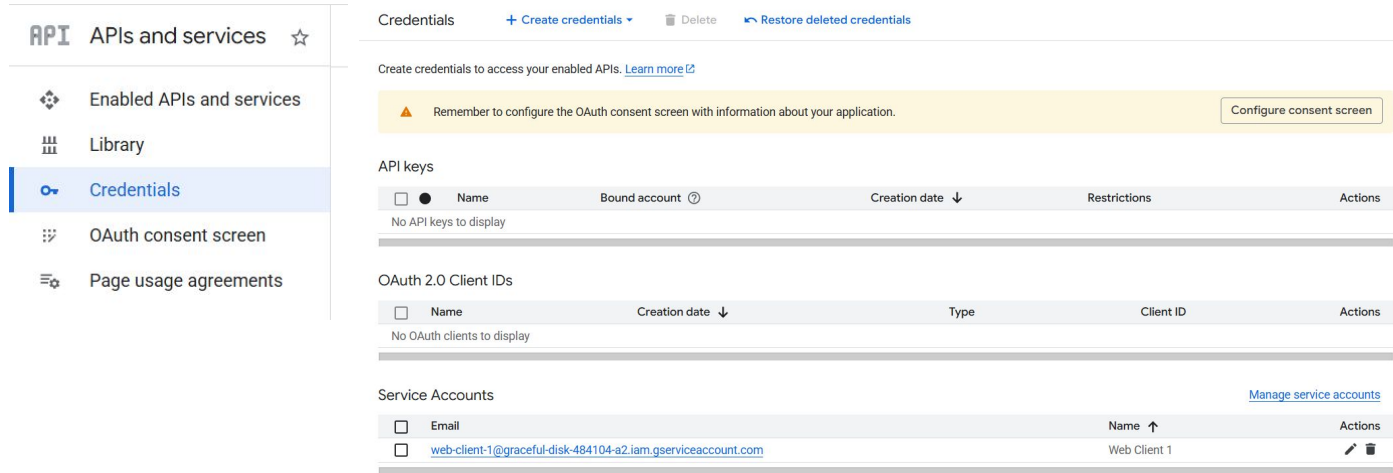
7. Select “Google Sheet API”
8. Click “Enable”



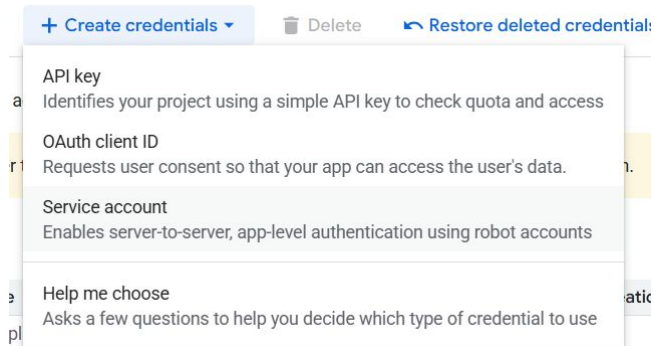
Data Collection - Reading Data from Google Sheet

Credentials Setting

1. Click Credentials and whole screen is displayed



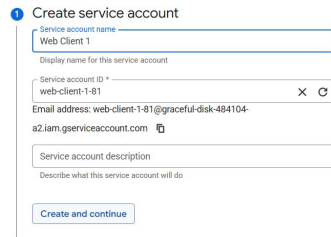
2. Click “Create credentials” and “Service Account”



Data Collection - Reading Data from Google Sheet

Credentials Setting

3. In "Service account name", enter "Web Client 1" and Click "Create and Continue"



1 Create service account

Service account name
Web Client 1

Display name for this service account

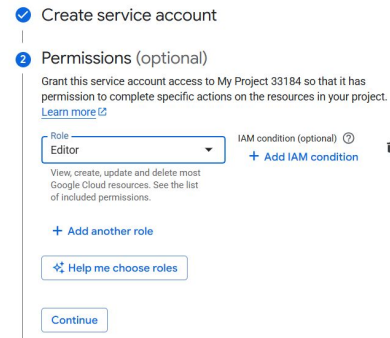
Service account ID *
web-client-1-81

Email address: web-client-1-81@graceful-disk-484104-a2.iam.gserviceaccount.com

Service account description
Describe what this service account will do

Create and continue

4. In "Select a role", enter "Editor", click "Continue"



2 Permissions (optional)

Grant this service account access to My Project 33184 so that it has permission to complete specific actions on the resources in your project.
[Learn more](#)

Role
Editor

IAM condition (optional) ⓘ
[+ Add IAM condition](#)

View, create, update and delete most Google Cloud resources. See the list of included permissions.

[+ Add another role](#)

[Help me choose roles](#)

Continue

5. Click "Done" to set credentials.

Service Accounts

[Manage service accounts](#)

☐	Email	Name ↑	Action
☐	web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com	Web Client 1	 

6. Copy the email like "web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com"

Data Collection - Reading Data from Google Sheet

Credentials Setting

8. Click “Service Account Email”

nts / Service account: 112433876246315493925

← Web Client 1

Details Permissions Keys Metrics Logs Principals with access

Service account details

Name
Web Client 1 Save

Description Save

9. Click “Keys”

Details Permissions **Keys** Metrics Logs Principals with access

10. Click “Add Key” and “Create New Key”

Add key ▾

Create new key

Upload existing key

11. Enter “JSON”,

Create private key for 'Web Client 1'

Downloads a file that contains the private key. Store the file securely because this key cannot be recovered if lost.

Key type

☒ JSON
Recommended

☐ P12
For backward compatibility with code using the P12 format

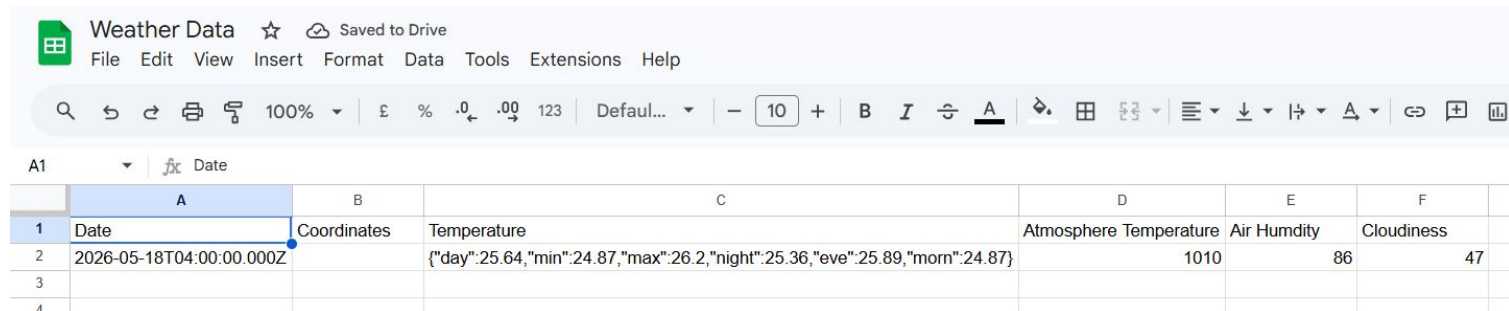
Cancel Create

12. A JSON Key is saved in your local PC

Data Collection - Reading Data from Google Sheet

Google Sheet Settings

1. Open previous weather data google sheet



The screenshot shows a Google Sheet titled "Weather Data" with a menu bar (File, Edit, View, Insert, Format, Data, Tools, Extensions, Help) and a toolbar. The spreadsheet contains the following data:

	A	B	C	D	E	F
1	Date	Coordinates	Temperature	Atmosphere Temperature	Air Humidity	Cloudiness
2	2026-05-18T04:00:00.000Z		{"day":25.64,"min":24.87,"max":26.2,"night":25.36,"eve":25.89,"morn":24.87}	1010	86	47
3						
4						

2. Click "File", Share, "Share with Others"
3. Enter "web-client-1@graceful-disk-484104-a2.iam.gserviceaccount.com"
4. Make sure it is in editor, click "Send"
5. From the google sheet website url like
<https://docs.google.com/spreadsheets/d/1BVdJkKIZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0/edit?gid=0#gid=0>
6. Copy sheet_id "**1BVdJkKIZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0**", range_name "**Sheet1!A1:F2**" in notepad

Data Collection - Reading Data from Google Sheet

Python Coding

1. Open Anaconda Prompt, enter “pip install google-api-core google-auth google-api-python-client google-auth-httpplib2 google-auth-oauthlib” to install google library

```
(base) C:\Users\Eddie Chow>pip install google-api-core google-auth google-api-python-client
Defaulting to user installation because normal site-packages is not writeable
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
WARNING: Ignoring invalid distribution -ransformers (c:\users\eddie chow\appdata\roaming\python\python39\site-packages)
Requirement already satisfied: google-api-core in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.34.1)
Requirement already satisfied: google-auth in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (1.35.0)
Requirement already satisfied: google-api-python-client in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (2.184.0)
Requirement already satisfied: googleapis-common-protos<2.0dev,>=1.56.2 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (1.70.0)
Requirement already satisfied: protobuf!=3.20.0,!<3.20.1,!<4.21.0,!<4.21.1,!<4.21.2,!<4.21.3,!<4.21.4,!<4.21.5,<4.0.0dev,>=3.19.5 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (3.20.3)
Requirement already satisfied: requests<3.0.0dev,>=2.18.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-core) (2.32.5)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (4.2.2)
Requirement already satisfied: pyasn1-modules>=0.2.1 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (0.2.8)
Requirement already satisfied: setuptools>=40.3.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (61.2.0)
Requirement already satisfied: six>=1.9.0 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (1.16.0)
Requirement already satisfied: rsa<5,>=3.1.4 in c:\programdata\anaconda3\lib\site-packages (from google-auth) (4.7.2)
Requirement already satisfied: httpplib2<1.0.0,>=0.19.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-python-client) (0.31.0)
Requirement already satisfied: google-auth-httpplib2<1.0.0,>=0.2.0 in c:\users\eddie chow\appdata\roaming\python\python39\site-packages (from google-api-python-client) (0.2.0)
```


Data Collection - Reading Data from Google Sheet

Python Coding

2. Open jupyter notebook
3. Type following code

```
import os
import csv
from google.oauth2.service_account import Credentials
from googleapiclient.discovery import build

SCOPES = ['https://www.googleapis.com/auth/spreadsheets']
SERVICES_ACCOUNT_FILE = "graceful-disk-484104-a2-152ebd9c692e.json"

credentials = Credentials.from_service_account_file(SERVICES_ACCOUNT_FILE,
scopes=SCOPES)
service = build('sheets', 'v4', credentials=credentials)

sheet_id = '1BVdJkKIZQPTdGrzL-h7ymTYT6unhYKgJ68vqqUYsjP0'
range_name = "Sheet1!A1:F2" # Specify the range with the sheet name
```

```
result = service.spreadsheets().values().get(spreadsheetId=sheet_id, range=range_name).execute()
values = result.get('values', [])
print(values)
```

```
2' en'' ,,,wru,, :zq' 8d' ,,wxX,, :ze' z' ,,uTdqf,, :z2' ze' ,,eLe,, :z2' 8d' ,,woxu,, :zq' 8d}', ' ,J0I0, ' ,8e, ' ,nA,]]  
[[,D9fe, ' ,C00tqtY9fge2, ' ,LewbeLefne, ' ,VfW0zbure6 LewbeLefne, ' ,Vti Httwtq7fl, ' ,CJoqq7ue22,] [ ,S0Se-02-J8I0n:00:00:0000,' , , '{,,q9λ,, :z
```

```
print('No data found.')
```

```
# Save to CSV file
```

```
writer = csv.writer(csv_file)
```

```
writer.writerows(values) # Write all the rows
```

```
print('Data has been saved to weatherforecastdata.csv')
```

Google Sheet Content Update

1. Open Anaconda Prompt, enter “`pip install gspread google-auth google-api-python-client`” to install google library
2. Open Jupyter notebook and type following python code

```
import gspread
from google.oauth2.service_account import Credentials
from googleapiclient.discovery import build

def update_values(sheet_id, range_name, value_input_option, values):
    SCOPES = ['https://www.googleapis.com/auth/spreadsheets']
    SERVICES_ACCOUNT_FILE = "graceful-disk-484104-a2-152ebd9c692e.json"

    credentials = Credentials.from_service_account_file(SERVICES_ACCOUNT_FILE,
                                                       scopes=SCOPES)
    service = build('sheets', 'v4', credentials=credentials)

    body = {"values": values}
```

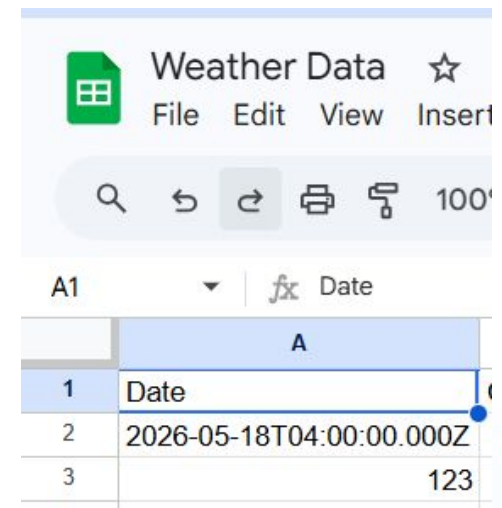
Google Sheet Content Update

```
# Update the range to just the starting cell for appending
result = service.spreadsheets().values().update(
    spreadsheetId=sheet_id,
    range=range_name, # Change range here
    valueInputOption=value_input_option,
    body=body
).execute()
```

```
print(f"{result.get('updatedCells')} cells updated.")
return result
```

```
sheet_id = '1mqrumbU1Si5aGmkg2jpffu7zn9uWxFvXX1Wnuwkas'
range_name = "Sheet1!A3"
value_input_option = 'USER_ENTERED'
values = [['123']]
```

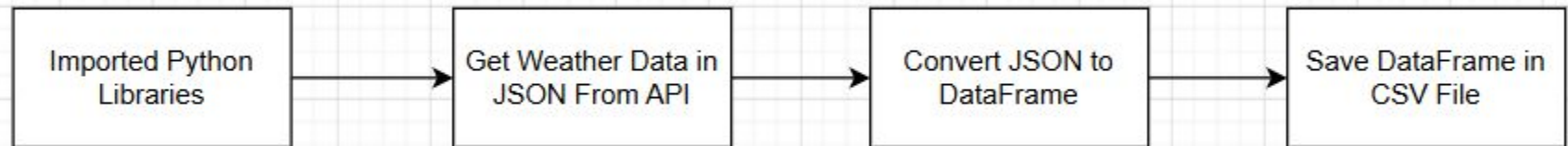
```
update_values(sheet_id, range_name, value_input_option, values)
```



Weather Data	
File	Edit View Insert
A1	
Date	
1	Date
2	2026-05-18T04:00:00.000Z
3	123

Data Collection - Reading Data from API

Below are the Flow Chart with the Process of Getting Weather Data from HKO API



Following Code contain 4 steps with the hypothesis

“Read Weather Forecasting Data from HKO API and Save in CSV File”

Step 1. Import the required libraries:

```
#import necessary library  
import requests  
import pandas as pd
```


Data Collection - Reading Data from API

Step 2: Load the weather forecasting data from ‘<https://data.weather.gov.hk/weatherAPI/opendata/>’

Before writing code, please take a look on API documentation at Hong Kong Observatory

Study the Weather API data dict Documentation in

https://www.hko.gov.hk/en/weatherAPI/doc/files/HKO_Open_Data_API_Documentation.pdf

1. Weather Information API

Dataset

- 9-day Weather Forecast
- Current Weather Report
- Local Weather Forecast
- Weather Warning Information
- Weather Warning Summary
- Special Weather Tips

API URL

<https://data.weather.gov.hk/weatherAPI/opendata/weather.php>

Please include valid parameters in API request. For valid parameters, please refer to Request table in this section.

HTTP Request Method

GET

Return Type

JSON

Request Example

<https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType=flw&lang=en>

Parameter

1. dataType
2. lang

Request

Parameter	Accepted values	Description
dataType	flw fnd rhrread warnsum warningInfo swt	flw: Local Weather Forecast fnd: 9-day Weather Forecast rhrread: Current Weather Report warnsum: Weather Warning Summary warningInfo: Weather Warning Information swt: Special Weather Tips
lang	en tc sc	en: English tc: Traditional Chinese sc: Simplified Chinese Default language: en

4

Data Collection - Reading Data from API

When browser load <https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType=flw&lang=en>



```
{
  "generalSituation": "The northeast monsoon is affecting the coast of Guangdong. Besides, a band of clouds is covering southern China.",
  "tcInfo": "",
  "fireDangerWarning": "",
  "forecastPeriod": "Weather forecast for this afternoon and tonight",
  "forecastDesc": "Mainly cloudy and dry. Bright periods in the afternoon. Moderate to fresh north to northeasterly winds.",
  "outlook": "Still cool in the morning in the next couple of days. The weather will improve gradually. Mainly fine in the middle and latter parts of this week to early next week. Warm and dry during the day. The temperature difference between day and night will be relatively large.",
  "updateTime": "2025-03-17T13:45:00+08:00"
}
```

generalSituation	tcInfo	fireDangerWarning	forecastPeriod	forecastDesc	outlook	updateTime
"The northeast monsoon is affecting the coast of Guangdong. Besides, a band of clouds is covering southern China"			Weather forecast for this afternoon and tonight	Mainly cloudy and dry. Bright periods in the afternoon. Moderate to fresh north to northeasterly winds	Still cool in the morning in the next couple of days. The weather will improve gradually. Mainly fine in the middle and latter parts of this week to early next week. Warm and dry during the day. The temperature difference between day and night will be relatively large.	2025-03-17T13:45:00+08:00

Data Collection - Reading Data from API

#Continue from previous page

Step 2: Load the weather forecasting data from 'https://data.weather.gov.hk/weatherAPI/opendata/'

```
# Initialize Parameters for the API
```

```
datatype = 'fnd'
```

```
lang = 'en'
```

```
apiurl = f'https://data.weather.gov.hk/weatherAPI/opendata/weather.php?dataType={datatype}&lang={lang}'
```

```
# Fetching the JSON data
```

```
response = requests.get(apiurl)
```

```
response.raise_for_status() # Check for request errors
```

```
# Converting JSON to a DataFrame
```

```
json_data = response.json()
```

Step 3: Converting JSON Data to a DataFrame

```
weatherforecast_df = pd.DataFrame(json_data['weatherForecast'])
```

```
# Display the DataFrame
```

```
print(weatherforecast_df.head())
```

```
forecastDate    week    forecastWind \
0  20250115  Wednesday    North force 4 to 5.
1  20250116  Thursday    North to northeast force 4, force 5 at first.
2  20250117  Friday    Northeast force 4, east force 5 later.
3  20250118  Saturday    East to northeast force 4 to 5, occasionally f...
4  20250119  Sunday    East to northeast force 2 to 3.

forecastWeather \
0  Mainly cloudy with a few light rain patches at...
1  Fine. Rather cool in the morning. Very dry dur...
2  Fine. Rather cool in the morning. Very dry dur...
3  Dry with sunny periods.
4  Mainly fine and dry.

forecastMaxtemp    forecastMintemp \
0  {'value': 20, 'unit': 'C'} {'value': 15, 'unit': 'C'}
1  {'value': 19, 'unit': 'C'} {'value': 13, 'unit': 'C'}
2  {'value': 19, 'unit': 'C'} {'value': 13, 'unit': 'C'}
3  {'value': 20, 'unit': 'C'} {'value': 15, 'unit': 'C'}
4  {'value': 21, 'unit': 'C'} {'value': 16, 'unit': 'C'}

forecastMaxrh    forecastMinrh \
0  {'value': 80, 'unit': 'percent'} {'value': 40, 'unit': 'percent'}
1  {'value': 65, 'unit': 'percent'} {'value': 35, 'unit': 'percent'}
2  {'value': 65, 'unit': 'percent'} {'value': 30, 'unit': 'percent'}
3  {'value': 70, 'unit': 'percent'} {'value': 40, 'unit': 'percent'}
4  {'value': 70, 'unit': 'percent'} {'value': 40, 'unit': 'percent'}

ForecastIcon  PSR
0          51  Low
1          92  Low
2          50  Low
3          51  Low
4          50  Low
```

Data Collection - Reading Data from API

#Continue from previous page

Step 4. Saving Weather Forecasting DataFrame to a CSV

```
csv_file = 'weatherForecast.csv' # Desired output CSV file
weatherforecast_df.to_csv(csv_file, index=False)
```

```
# Output successful message for saving weather forecasting data.
print(f'Data has been successfully saved to {csv_file}')
```

A	B	C	D	E	F	G	H	I	J
forecastDate	week	forecastWind	forecastWeather	forecastMaxtemp	forecastMintemp	forecastMaxrh	forecastMinrh	ForecastIcon	PSR
20250115	Wednesday	North force 4 to 5.	Mainly cloudy with a few	{'value': 20, 'unit':	{'value': 15, 'unit':	{'value': 80, 'un	{'value': 40, 'un	51	Low
20250116	Thursday	North to northeast force	Fine. Rather cool in the r	{'value': 19, 'unit':	{'value': 13, 'unit':	{'value': 65, 'un	{'value': 35, 'un	92	Low
20250117	Friday	Northeast force 4, east	Fine. Rather cool in the r	{'value': 19, 'unit':	{'value': 13, 'unit':	{'value': 65, 'un	{'value': 30, 'un	50	Low
20250118	Saturday	East to northeast force	Dry with sunny periods.	{'value': 20, 'unit':	{'value': 15, 'unit':	{'value': 70, 'un	{'value': 40, 'un	51	Low
20250119	Sunday	East to northeast force	Mainly fine and dry.	{'value': 21, 'unit':	{'value': 16, 'unit':	{'value': 70, 'un	{'value': 40, 'un	50	Low
20250120	Monday	East to northeast force	Fine and dry.	{'value': 22, 'unit':	{'value': 16, 'unit':	{'value': 70, 'un	{'value': 40, 'un	50	Low
20250121	Tuesday	East to northeast force	Fine and dry.	{'value': 22, 'unit':	{'value': 17, 'unit':	{'value': 70, 'un	{'value': 40, 'un	50	Low
20250122	Wednesday	East force 4, occasional	Sunny periods.	{'value': 21, 'unit':	{'value': 17, 'unit':	{'value': 75, 'un	{'value': 50, 'un	51	Low
20250123	Thursday	East force 4, occasional	Sunny intervals.	{'value': 21, 'unit':	{'value': 17, 'unit':	{'value': 75, 'un	{'value': 50, 'un	52	Low

Definition

- a web crawler is a computer program that browses the World Wide Web in a methodical, automated manner. (Wikipedia)

Utilities:

- Gather pages from the Web.
- Support a search engine, perform data mining and so on.

Object:

- Text, video, image and so on.
- Link structure

Web Features of a Crawler

Should provide:

- Distributed
- Scalable
- Performance and efficiency
- Quality
- Freshness
- Extensible

Some scraping knowledge

1. HTTP: the communication protocol
2. HTML: the language in which web pages are defined
3. JS: javascript (code executing in the browser)
4. CSS: style sheets, how web pages are styled. Important, but does not contain data.
5. JPG, PNG, BMP: images, usually not interesting
6. CSV / TXT / JSON / XML: data, interesting !!!

Obtaining Data

Data can come from:

- You curate it
- Someone else provides it, all pre-packaged for you (e.g., files)
- Someone else provides an API
- Someone else has available content, and you try to take it (web scraping)

Obtaining Data – Web Scraping

Web Scraping

- Using programs to get data from online
- Often much faster than manually copying data!
- Transfer the data into a form that is compatible with your code

Obtaining Data – Web Scrapping

Why scrapping the web?

- Vast source of information; can combine with multiple datasets
- Companies have not provided APIs
- Automate tasks
- Keep up with sites / real-time data
- Fun!

Robots.txt

- Specified by web site owner
- Gives instructions to web robots (e.g., your code)
- Located at the top-level directory of the web server
 - E.g., <http://google.com/robots.txt>

Robots.txt

- Protocol for giving spiders ("robots") limited access to a website

www.robotstxt.org/wc/norobots.html

- Website announced its request on what can(not) be crawled
 - For a server, create a file /robots.txt
 - This file specifies access restrictions

Robots.txt

- Protocol for giving spiders ("robots") limited access to a website

www.robotstxt.org/wc/norobots.html

- Website announced its request on what can(not) be crawled
 - For a server, create a file /robots.txt
 - This file specifies access restrictions

An example of robots.txt

- No rebot should visit any URL starting with "/yoursite/temp/", except the rebot called "searchengine":
- User-agent: *
- Disallow: /yoursite/temp/
- User-agent: searchengine
- Disallow:

Web Servers

- A server maintains a long-running process (also called a daemon), which listens on a pre-specified port
- It responds to requests, which is sent using a protocol called HTTP (HTTPS is secure)
- Our browser sends these requests and downloads the content, then displays it
- 2– request was successful, 4– client error, often `page not found`; 5– server error (often that your request was incorrectly formed)

Obtaining Data – Web Scraping

HTML

- Tags are denoted by angled brackets
- Almost all tags are in pairs e.g.,
`<p>Hello</p>`
- Some tags do not have a closing tag e.g.,
`<br/>`

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Title</title>
  </head>
  <body>
    <h1>Body Title</h1>
    <p>Body Content</p>
  </body>
</html>
```

Obtaining Data – Web Scrapping

HTML

- `<html>`, indicates the start of an html page
- `<body>`, contains the items on the actual webpage (text, links, images, etc)
- `<p>`, the paragraph tag. Can contain text and links
- `<a>`, the link tag. Contains a link url, and possibly a description of the link
- `<input>`, a form input tag. Used for text boxes, and other user input
- `<form>`, a form start tag, to indicate the start of a form
- `<img>`, an image tag containing the link to an image

Obtaining Data – Web Scraping

How to Web scrape:

1. Get the webpage content

- Requests (Python library) gets a webpage for you

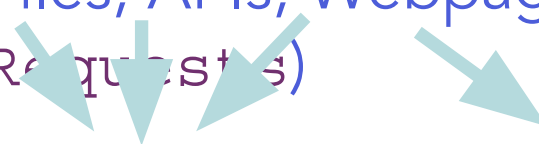
2. Parse the webpage content

- (e.g., find all the text or all the links on a page)
- BeautifulSoup (Python library) helps you parse the webpage.
- Documentation:

<http://crummy.com/software/BeautifulSoup>

The Big Picture Recap

Data Sources Files, APIs, Webpages (via Requests)



Data Parsing Regular Expressions, BeautifulSoup

Data Structures/Storage Traditional lists/dictionaries, PANDAS

Models Linear Regression, Logistic Regression, kNN, etc

BeautifulSoup only concerns webpage data

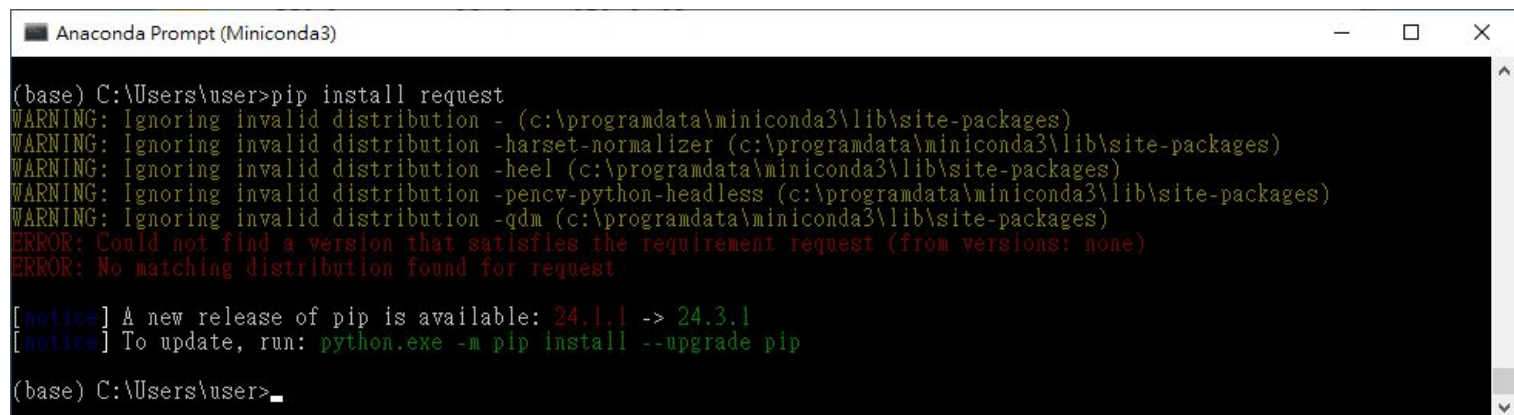
Obtaining Data – Web Scraping

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



```
Anaconda Prompt (Miniconda3)
(base) C:\Users\user>
```

2. Install python package requests by entering “pip install requests”



```
Anaconda Prompt (Miniconda3)
(base) C:\Users\user>pip install request
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -qdm (c:\programdata\miniconda3\lib\site-packages)
ERROR: Could not find a version that satisfies the requirement request (from versions: none)
ERROR: No matching distribution found for request

[notice] A new release of pip is available: 24.1.1 -> 24.3.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(base) C:\Users\user>_
```

Obtaining Data – Web Scraping

3. Install python package beautifulsoup4 by entering “pip install beautifulsoup4”

```
Anaconda Prompt (Miniconda3)

(base) C:\Users\user>pip install beautifulsoup4
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -qdm (c:\programdata\miniconda3\lib\site-packages)
Requirement already satisfied: beautifulsoup4 in c:\programdata\miniconda3\lib\site-packages (4.12.3)
Requirement already satisfied: soupsieve>1.2 in c:\programdata\miniconda3\lib\site-packages (from beautifulsoup4) (2.3.2.post1)
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
```

4. Enter “python” in the command

```
Anaconda Prompt (Miniconda3) - python

(base) C:\Users\user>python
Python 3.9.12 (main, Apr 4 2022, 05:22:27) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> _
```


Obtaining Data – Web Scraping

1. Get the webpage content

Requests (Python library) gets a webpage for you

```
import requests
```

```
url = "https://www.nytimes.com/international/"
```

```
page = requests.get(url)
```

```
page.status_code
```

```
page.content
```

Obtaining Data – Web Scrapping

1. Get the webpage content

Requests (Python library) gets a webpage for you

```
import requests
```

```
url = "https://www.nytimes.com/international/"
```

```
page = requests.get(url)
```

```
page.status_code
```

```
page.content
```

Gets the status from the webpage request.
200 means success.
404 means page not found.

Obtaining Data – Web Scraping

1. Get the webpage content

Requests (Python library) gets a webpage for you

```
import requests
```

```
url = "https://www.nytimes.com/international/"
```

```
page = requests.get(url)
```

```
page.status_code
```

```
page.content
```

Returns the content of the response, in bytes.

Obtaining Data – Web Scraping

2. Parse the webpage content

BeautifulSoup (Python library) helps you parse a webpage

```
from bs4 import BeautifulSoup

soup = BeautifulSoup(page.content,
    "html.parser")

soup.title
soup.title.text
```

Obtaining Data – Web Scraping

2. Parse the webpage content

BeautifulSoup (Python library) helps you parse a webpage

```
from bs4 import BeautifulSoup
```

```
soup = BeautifulSoup(page.content,  
"html.parser")
```

```
soup.title
```

```
soup.title.text
```

Returns the full context, including the title tag. e.g.,

```
<title data-rh="true">The New York  
Times - Breaking News</title>
```

Obtaining Data – Web Scraping

2. Parse the webpage content

BeautifulSoup (Python library) helps you parse a webpage

```
soup = BeautifulSoup(page.content,  
"html.parser")
```

```
soup.title
```

```
soup.title.text
```

Returns the text part of the title tag. e.g.,

The New York Times - Breaking
News

Obtaining Data – Web Scraping

BeautifulSoup

- Helps make messy HTML digestible
- Provides functions for quickly accessing certain sections of HTML content

Exempl

```
import bs4
## get bs4 object
soup = bs4.BeautifulSoup(source)
## all a tags
soup.findAll('a')
## first a
soup.find('a')
## get all links in the page
link_list = [l.get('href') for l in soup.findAll('a')]
```

Obtaining Data – Web Scraping

HTML is a tree

- You don't have to access the HTML as a tree, though;
- Can immediately search for tags/content of interest (a la previous slide)

Exempl

```
tree = bs4.BeautifulSoup(source)

## get html root node
root_node = tree.html

## get head from root using contents
head = root_node.contents[0]

## get body from root
body = root_node.contents[1]

## could directly access body
tree.body
```

Obtaining Data – Web Scraping

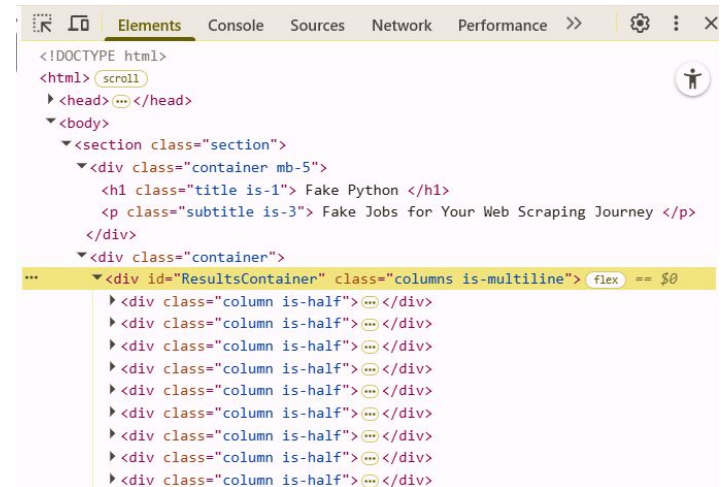
```
import requests
from bs4 import BeautifulSoup
```

```
URL = "https://realpython.github.io/fake-jobs/"
page = requests.get(URL)
```

```
soup = BeautifulSoup(page.content, "html.parser")
```

Fake Python

Fake Jobs for Your Web Scraping Journey



**Senior Python Developer**
Payne, Roberts and Davis

Stewartbury, AA
2021-04-08

[Learn](#)[Apply](#)

**Energy engineer**
Vasquez-Davidson

Christopherville, AA
2021-04-08

[Learn](#)[Apply](#)

**Legal executive**
Jackson, Chambers and Levy

Port Ericaburgh, AA
2021-04-08

[Learn](#)[Apply](#)

**Fitness centre manager**
Savage-Bradley

East Seanview, AP
2021-04-08

[Learn](#)[Apply](#)

Obtaining Data – Web Scraping

```
URL = "https://realpython.github.io/fake-jobs/"  
page = requests.get(URL)
```

```
soup = BeautifulSoup(page.content, "html.parser")
```

```
results = soup.find(id="ResultsContainer")  
print(results.prettify())
```

```
>>> print(results.prettify())  
<div class="columns is-multiline" id="ResultsContainer">  
  <div class="column is-half">  
    <div class="card">  
      <div class="card-content">  
        <div class="media">  
          <div class="media-left">  
            <figure class="image is-48x48">  
                
            </figure>  
          </div>  
          <div class="media-content">  
            <h2 class="title is-5">  
              Senior Python Developer  
            </h2>  
            <h3 class="subtitle is-6 company">  
              Payne, Roberts and Davis  
            </h3>  
          </div>  
        </div>  
      </div>  
    </div>  
  </div>
```

Obtaining Data – Web Scraping

Find Elements by HTML Class Name

You've seen that every job posting is wrapped in a `<div>` element with the class `card-content`. Now you can work with your new object called `results` and select only the job postings in it. These are, after all, the parts of the HTML that you're interested in! You can pick out all job cards in a single line of code:

Python

```
>>> job_cards = results.find_all("div", class_="card-content")
```

Here, you call `.find_all()` on `results`, which is a BeautifulSoup object. It returns an `iterable` containing all the HTML for all the job listings displayed on that page.

Take a look at all of them:

Python

```
>>> for job_card in job_cards:
...     print(job_card, end="\n" * 2)
...
```

```
<div class="card-content">
<div class="media">
<div class="media-left">
<figure class="image is-48x48">

<h2 class="title is-5">Senior Python Developer</h2>
<h3 class="subtitle is-6 company">Payne, Roberts and Davis</h3>
</div>
</div>
<div class="content">
<p class="location">
    Stewartbury, AA
</p>
<p class="is-small has-text-grey">
<time datetime="2021-04-08">2021-04-08</time>
```

Extract title Content

```
>>> for job_card in job_cards:
...     title_element = job_card.find("h2", class_="title")
...     print(title_element)
...
<h2 class="title is-5">Senior Python Developer</h2>
<h2 class="title is-5">Energy engineer</h2>
<h2 class="title is-5">Legal executive</h2>
<h2 class="title is-5">Fitness centre manager</h2>
<h2 class="title is-5">Product manager</h2>
<h2 class="title is-5">Medical technical officer</h2>
<h2 class="title is-5">Physiological scientist</h2>
<h2 class="title is-5">Textile designer</h2>
<h2 class="title is-5">Television floor manager</h2>
<h2 class="title is-5">Waste management officer</h2>
<h2 class="title is-5">Software Engineer (Python)</h2>
<h2 class="title is-5">Interpreter</h2>
<h2 class="title is-5">Architect</h2>
<h2 class="title is-5">Meteorologist</h2>
```

Extract Content of title, company, location

Python

```
>>> for job_card in job_cards:
...     title_element = job_card.find("h2", class_="title")
...     company_element = job_card.find("h3", class_="company")
...     location_element = job_card.find("p", class_="location")
...     print(title_element)
...     print(company_element)
...     print(location_element)
...     print()
...
<h2 class="title is-5">Senior Python Developer</h2>
<h3 class="subtitle is-6 company">Payne, Roberts and Davis</h3>
<p class="location">
    Stewartbury, AA
</p>

<!-- ... -->
```

Full Article : <https://realpython.com/beautiful-soup-web-scraper-python/>

Obtaining Data – Web Scraping

Open Browser and enter
<https://www.hk01.com/>

Right Click and Select “Inspect”



Select the Content that you want to extract

Obtaining Data – Web Scraping

Right Click and click "Edit As HTML"
Copy part of the code until `</div>`

The screenshot shows a web browser window with a news article from HKU SPACE. The article is titled "李嘉誠治肝癌新儀器料歸三私院 陸志聰：公院引入須顧人手及預算" and is dated "2 小時前". The article content includes a video player and a text block about "Plug & Play Installation in under 4 Hours." and "Reducing TCO Total Cost of Ownership by 44%.".

On the right side of the browser window, a right-click context menu is open, showing various options. The option "Edit As HTML" is highlighted, indicating the user's intention to view the page's source code for web scraping.

The background of the browser window shows a news article with a video player and a text block about "Plug & Play Installation in under 4 Hours." and "Reducing TCO Total Cost of Ownership by 44%.".

Obtaining Data – Web Scraping

Open Perplexity at <https://www.perplexity.ai/>

Enter prompt "write python beautifulsoap to extract data from website url <https://www.hk01.com/> with html code {htmlcode} "

Where {htmlcode} = "<div class="page__layout-navbar" data-testid="page-layout-navbar"><div class="flex flex-none items-center md:hidden"><button data-testid="hamburger-menu-button" class="box-border w-6 px-0 py-3 m-0 transition-all duration-[0.6s] ease-[cubic-bezier(0.19,1,0.22,1)] [transform:translateZ(0)] border-none outline-none rounded-none cursor-pointer shadow-none bg-transparent focus:shadow-none focus:outline-none" aria-label="mobile-menu"></button></div>"

Obtaining Data – Web Scraping

In original perplexity prompt, type the following prompt
"Add code that save it into csv file"

A	
1	text,url,class
2	./,
3	港聞,/zone/1/%E6%B8%AF%E8%81%9E,
4	兩會焦點,/issue/10355/%E5%85%A9%E6%9C%832025-%E5%9C%8B%E9%9A%9B%E5%B1%80%
5	娛樂,/zone/2/%E5%A8%9B%E6%A8%82,
6	最平酒店,https://clk.omgt3.com/?AID=2200313&PID=55029&uid=content&uid2=channel_旅遊_editor
7	國際,/zone/4/%E5%9C%8B%E9%9A%9B,
8	即時,/latest,
9	熱榜,/hot,
10	生活,/zone/8/%E7%94%9F%E6%B4%BB,
11	科技,/zone/11/%E7%A7%91%E6%8A%80%E7%8E%A9%E7%89%A9,
12	中國,/zone/5/%E4%B8%AD%E5%9C%8B,
13	體育,/zone/3/%E9%AB%94%E8%82%B2,
14	01深圳,/zone/25/01%E6%B7%B1%E5%9C%B3,
15	經濟,/zone/14/%E7%B6%93%E6%BF%9F,
16	觀點,/zone/12/%E8%A7%80%E9%BB%9E,
17	健康,/zone/24/%E5%81%A5%E5%BA%B7,
18	好食玩飛,/zone/19/%E5%A5%BD%E9%A3%9F%E7%8E%A9%E9%A3%9B,
19	女生,/zone/6/%E5%A5%B3%E7%94%9F,
20	熱話,/zone/7/%E7%86%B1%E8%A9%B1,
21	藝文格物,/zone/9/%E8%97%9D%E6%96%87%E6%A0%BC%E7%89%A9,
22	社區,/zone/10/%E7%A4%BE%E5%8D%80,
23	教育,/zone/23/%E6%95%99%E8%82%B2,
24	簡,/,"font-light, pointer-events-none, !text-light-n5"
25	
26	
27	Mobile Menu Details:
28	Exists: True

How To Do Data Mining: Machine Learning Techniques

Features / Attributes

1. District
2. Floor

Outcome

1. Price Per Square

<https://www.28hse.com/en/rent/apartment/property-3496859>

Floor zone

Middle Floor

Saleable Area

807 ft²

Unit Price: @46.8

<https://www.28hse.com/en/rent/apartment>



♥BEST CHOICE♥ \$37800 Front line sea view...

Tsuen Wan Ocean Pride | Unit C, Mid Floor, Tower 1, Ocean Supreme, Middle Floor 12 minutes ago posted

Saleable Area: 807 ft² @46.8

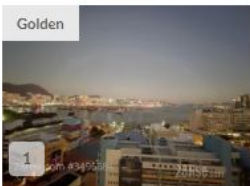
美聯物業

Apartment Good view Sea view Elegant

Air conditioner Washing Machine Water Heaters Refrigerator

Induction Hob Cooker Hood

Lease HKD\$37,800



The largest room with sea view•Rarely availa...

Hung Hom Upper East | High Floor, Tower 1A 2 hours ago posted

Saleable Area: 375 ft² @49.3

Huicheng Property Company Limited

1 Bedroom, 1 Bathroom Western Style Bldg

Stand-alone Building Good view Sea view Elegant Microwave Oven

Air conditioner Washing Machine Water Heaters

Lease HKD\$18,500



3-bedroom Western-style building with open ...

Kwun Tong Hamden Court | High Floor, Block A 3 hours ago posted

Saleable Area: 481 ft² @35.8

Kam Lung Property Agency Co.

3 Bedrooms, 1 Bathroom Western Style Bldg Good view

Elegant

Lease HKD\$17,200

How To Do Data Mining: Machine Learning Techniques

In perplexity, enter the prompt “write python beautifulsoap to extract data from website url and save data into csv file using {pythoncode} from original html code {htmlcode}”

```
pythoncode = ""
import csv
import pandas as pd
from bs4 import BeautifulSoup
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.firefox.options import Options
import requests
import time

from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

def main(url:str, per_page:int = 120, **kwargs):
    options = kwargs.get('driver_options', Options())
    # per_page = kwargs.get('per_page', 120)
    # url = f"{url}?per_page={per_page}"
    options.headless = True

    firefox_options = Options()
    firefox_options.set_preference("dom.webdriver.enabled", False)

    driver = webdriver.Firefox(
        options=firefox_options
    )

    driver.get(url)
    # Get the page source
    html = driver.page_source

    # Create a BeautifulSoup object from the HTML
    soup = BeautifulSoup(html, 'html.parser')
```

How To Do Data Mining: Machine Learning Techniques

```
htmlcode = ""
<div class="ui segment search_results_div" id="search_results_div">    <div class="listItems ui divided
items">
    <div class="item property_item ">
        <div class="image myimage desktop_myimage">
            <a class="detail_page" href="https://www.28hse.com/en/rent/office/property-3445279"
attr1="3445279" target="_blank">
                

            <div class="myimage_count_outer"><div class="ui labels myimage_count">
                <span class="ui label" style="">
                    10

            </span></div></div>

            <div class="ui top left attached small label
grade_label">Golden</div>

        </a>

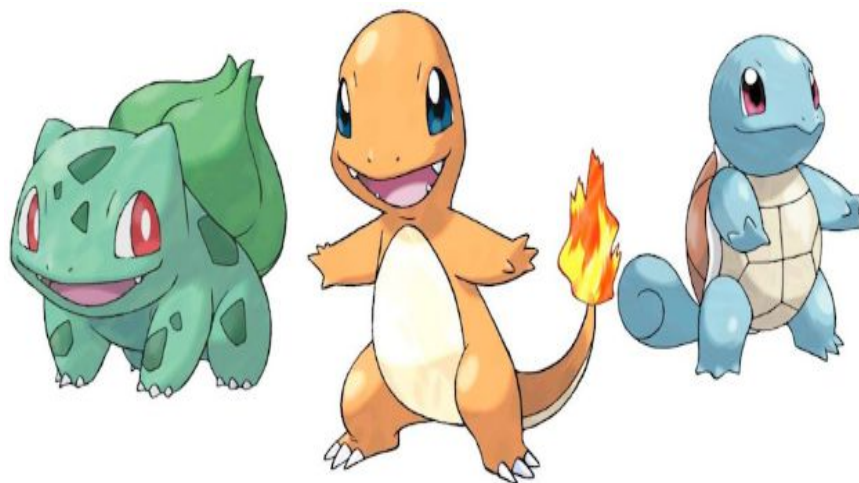
    </div>
    <div class="content">
        <div class="right floated">
            <div class="fav" attr1="3445279" attr2="inactive"><i class="ui red heart
large outline icon"></i></div>
        </div>
    </div>
</div>
```


Download the data

Download the Pokemon dataset from:

https://github.com/innoviai/IPA_Courses-May2026/blob/main/pokemon_dataset.csv

Save the pokemon dataset file in a location you'll remember.



Data Collection – Fetching Data using Pandas

What / Why?

- Pandas is an *open-source* Python library (anyone can contribute)
- Allows for high-performance, easy-to-use data structures and data analysis
- Unlike NumPy library which provides multi-dimensional arrays, Pandas provides 2D table object called **DataFrame** (akin to a spreadsheet with column names and row labels).
- Used by *a lot* of people

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How

- import **pandas** library (convenient to rename it)
- Use **read_csv()** function

```
import pandas as pd  
dataframe = pd.read_csv("yourfile.csv")
```

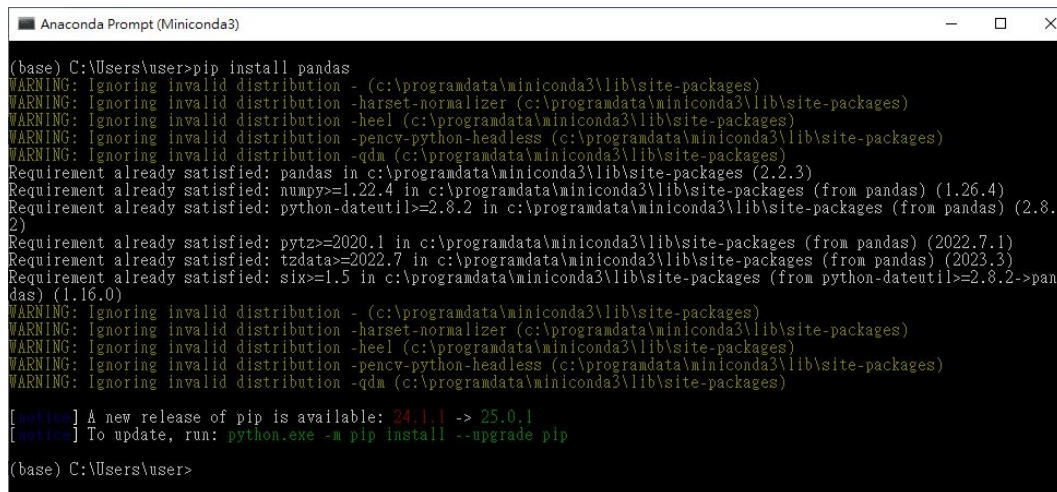
Library Installation in Python

1. Open tools “Anaconda Prompt” which will direct you to the command prompt.



```
Anaconda Prompt (Miniconda3)
(base) C:\Users\user>
```

2. Install python package requests by entering “pip install pandas”



```
Anaconda Prompt (Miniconda3)
(base) C:\Users\user>pip install pandas
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -qdm (c:\programdata\miniconda3\lib\site-packages)
Requirement already satisfied: pandas in c:\programdata\miniconda3\lib\site-packages (2.2.3)
Requirement already satisfied: numpy>=1.22.4 in c:\programdata\miniconda3\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\miniconda3\lib\site-packages (from pandas) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\miniconda3\lib\site-packages (from pandas) (2022.7.1)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\miniconda3\lib\site-packages (from pandas) (2023.3)
Requirement already satisfied: six>=1.5 in c:\programdata\miniconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
WARNING: Ignoring invalid distribution - (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -harset-normalizer (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -heel (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -pencv-python-headless (c:\programdata\miniconda3\lib\site-packages)
WARNING: Ignoring invalid distribution -qdm (c:\programdata\miniconda3\lib\site-packages)
[notice] A new release of pip is available: 24.1.1 -> 25.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(base) C:\Users\user>
```

Reading in the data

First we import the Python packages we are going to use.
Then we use Pandas to load in the dataset as a data frame.

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
import seaborn as sns
```

```
df1 = pd.read_csv("pokemon_dataset.csv", index_col=0, encoding="ISO-8859-1")
```

NOTE: The argument `index_col` argument states that we'll treat the first column of the dataset as the ID column.

NOTE: The encoding argument allows us to by pass an input error created by special characters in the data set.

Examine the data set

```
print(df1.head())
```

	Name	Type 1	Type 2	Total	...	Sp. Def	Speed	Stage	Legendary
#					...				
1	Bulbasaur	Grass	Poison	318	...	65	45	1	False
2	Ivysaur	Grass	Poison	405	...	80	60	2	False
3	Venusaur	Grass	Poison	525	...	100	80	3	False
4	Charmander	Fire	NaN	309	...	50	65	1	False
5	Charmeleon	Fire	NaN	405	...	65	80	2	False

```
print(df1.describe())
```

	Total	HP	Attack	...	Sp. Def	Speed	Stage
count	151.00000	151.00000	151.00000	...	151.00000	151.00000	151.00000
mean	407.07947	64.211921	72.549669	...	66.019868	68.933775	1.582781
std	99.74384	28.590117	26.596162	...	24.197926	26.746880	0.676832
min	195.00000	10.000000	5.000000	...	20.000000	15.000000	1.000000
25%	320.00000	45.000000	51.000000	...	49.000000	46.500000	1.000000
50%	405.00000	60.000000	70.000000	...	65.000000	70.000000	1.000000
75%	490.00000	80.000000	90.000000	...	80.000000	90.000000	2.000000
max	680.00000	250.00000	134.00000	...	125.00000	140.00000	3.000000

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Common Panda functions

High-level viewing:

- `head()` – first N observations
- `tail()` – last N observations
- `columns()` – names of the columns
- `describe()` – statistics of the quantitative data
- `dtypes()` – the data types of the columns

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Common Panda functions

Accessing/processing:

- `df["column_name"]`
- `Df.column_name`
- `.max()`, `.min()`, `.idxmax()`, `.idxmin()`
- `<dataframe> <conditional statement>`
- `.loc[]` – label-based accessing
- `.iloc[]` – index-based accessing
- `.sort_values()`
- `.isnull()`, `.notnull()`



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THANK YOU

